

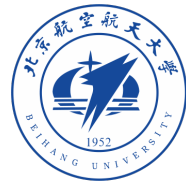
The 33rd International Symposium on Graph Drawing and Network Visualization

(invited TVCG talk) GD'25

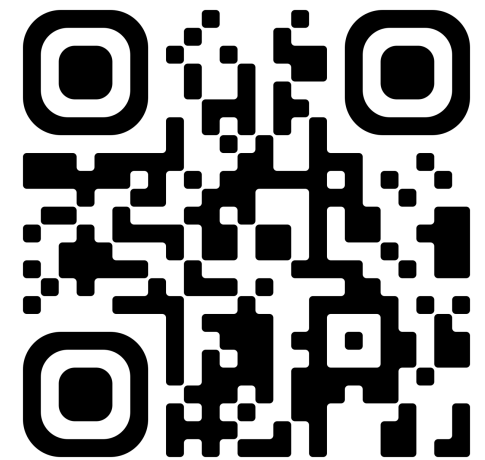
GeneticPrism: Multifaceted Visualization of Citation-based Scholarly Research Evolution

Ye Sun, Zipeng Liu, Yuankai Luo, Lei Xia, Lei Shi*

Beihang University, Beijing, China

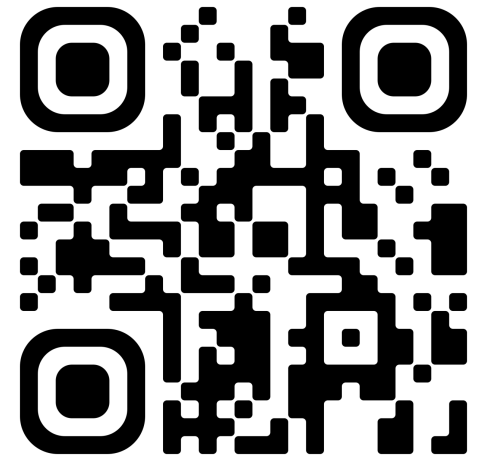
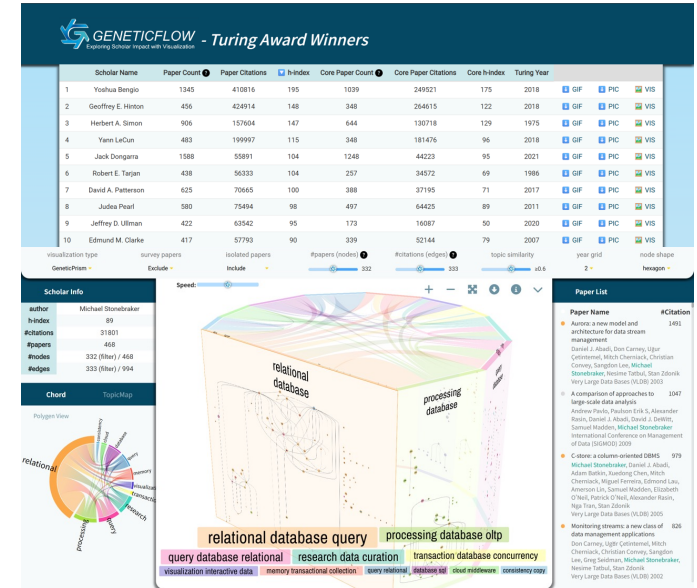


北京航空航天大学
BEIHANG UNIVERSITY



<https://genetic-flow.com>

System Interface



29,541 users from 76 countries and 209,895 visits since 2023-09.

<https://genetic-flow.com>

Background

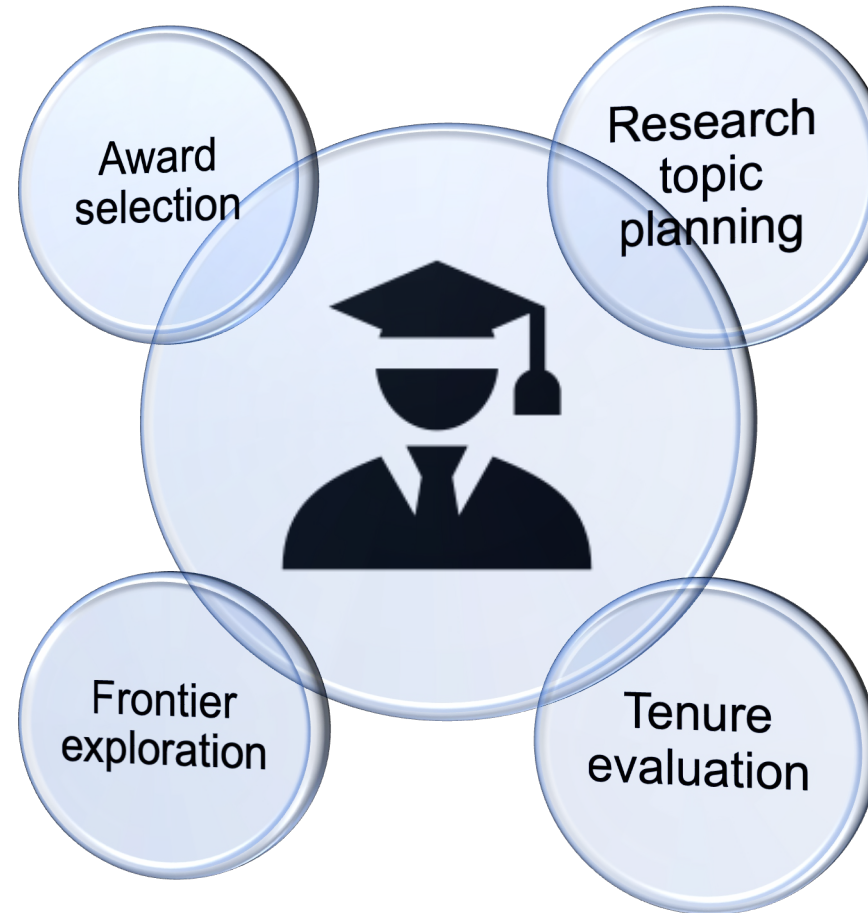
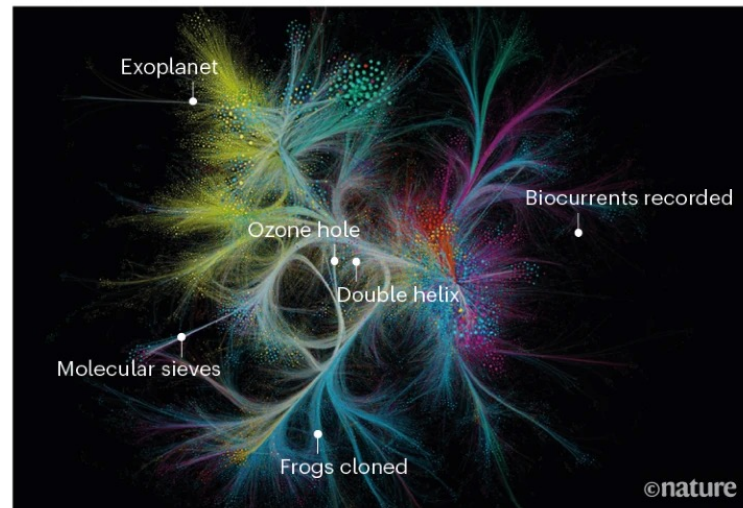
Analyzing evolution of a scholar's scientific research is used for:

CO-CITATION NETWORK

Each Nature paper is a dot. Dots are linked if another paper cites both. Some articles (colourful clusters) are cited by many disciplines, others (monotone areas) are deeply embedded in their own disciplines. (See go.nature.com/n150int for an interactive version, including references to the six highlighted papers.)

Discipline

- Arts
- Biology
- Biomedical research
- Chemistry
- Clinical medicine
- Earth and space
- Engineering and technology
- Health
- Humanities
- Mathematics
- Physics
- Business and management
- Psychology
- Social sciences



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Related Work: Scholar Indicator & Publication List

- Good for **ranking** but not for scholar profile reasoning and understanding



Daniel Keim
Professor of Computer Science, Data Analysis and Visualization, [University of Konstanz](#)
Verified email at uni-konstanz.de - [Homepage](#)
[Data Analysis](#) [Information Visualization](#) [Visual Analytics](#) [Visualization](#) [Clustering](#)

FOLLOWING

TITLE

[On the surprising behavior of distance metrics in high dimensional data](#)
CC Aggarwal, A Hinneburg, DA Keim
International conference on database theory, 420-434

[Information visualization and visual data mining](#)
DA Keim
IEEE transactions on Visualization and Computer Graphics 8 (1), 1-8

[The X-tree: An index structure for high-dimensional data](#)
S Berchtold, DA Keim, HP Kriegel

[An efficient approach to clustering in large multimedia databases](#)
A Hinneburg, DA Keim
Bibliothek der Universität Konstanz 98, 58-65

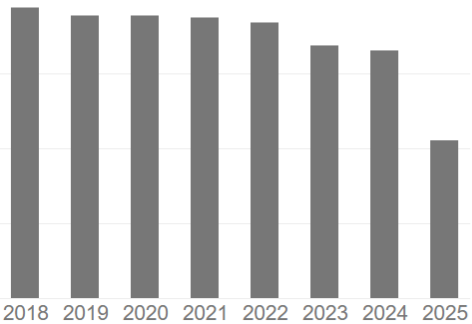
[Visual analytics: Definition, process, and challenges](#)
D Keim, G Andrienko, JD Fekete, C Görg, J Kohlhammer, G Melançon
Information visualization: Human-centered issues and perspectives, 1-10

[Searching in high-dimensional spaces: Index structures for multimedia databases](#)
C Böhm, S Berchtold, DA Keim
ACM Computing Surveys (CSUR) 33 (3), 322-373

Cited by

VIEW ALL

	All	Since 2020
Citations	50024	15534
h-index	95	57
i10-index	436	253



Year	Citations
2018	3100
2019	3100
2020	3100
2021	3100
2022	3100
2023	2325
2024	2325
2025	1550

Search a VIS author

Search

Ben Shneiderman 100+ 100+ 20+

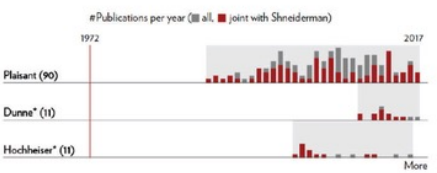


1972-2017: Shneiderman's #pub. per year (all, joint with Plaisant)

Ben Shneiderman is an active and longtime research contributor with more than **400 publications** since 1972. Shneiderman's published work includes **206 journal articles** and **197 proceedings papers**.

Shneiderman is a current core member of the **visualization** community. Shneiderman's expertise mainly covers the subfields of **information visualization** and **visual analytics**. A focus area of the author is the visualization of hierarchy data. Other current research areas of the author are **human-computer interaction** and **bioinformatics**. Researchers with similar areas of expertise are Stuart K. Card, George G. Robertson, Raimund Dachselt, and Pourang Irani.

Shneiderman's most frequent co-author and past supervisee is **Catherine Plaisant**. It is a prolonged and still ongoing collaboration since 1988 with **90 publications**. Cody Dunne* is the second most frequent co-author, a collaboration that produced **11 publications** between 2009 and 2015. Together with Harry Hochheiser*, the author published **11 research papers** between 2000 and 2011. In addition to Plaisant, further supervisees of Shneiderman with considerable amount of publications are Hyunmo Kang, Taowei David Wang, Adam Perer, Krist Wongsuphasawat, and Awan Sopan. Shneiderman's supervisees Catherine Plaisant, Jinwook Seo, and Chris North are already supervising other researchers. Regarding collaboration subgroups, Shneiderman has worked with Anne Rose* and Catherine Plaisant on human-computer interaction and produced **7 research papers**. Another notable group is with Marc A. Smith* and Derek L. Hansen* resulting in **5 publications**.



#Publications per year (all, joint with Shneiderman)

Mutual Publications (9): Shneiderman and Plaisant, 2013

30 years at the University of Maryland's human-computer interaction lab (HCIL)

Ben Shneiderman, Kent L. Norman*, Catherine Plaisant, Benjamin B. Bederson, Allison Drubin*, Jennifer Golbeck*, Interactions, 2013

human-computer interaction

An Information-Centric Framework for Designing Patient-Centered Medical Decision Aids and Risk Communication

Lyndsey Franklin, Catherine Plaisant, Ben Shneiderman

AMIA 2013, American Medical Informatics Association Annual Symposium, Washington, DC, USA, November 16-20, 2013

bioinformatics

Exploring Data Distributions: Visual Design and Evaluation

Awan Sopan, Manuel Fraine*, Meirav Tsaib-Maimon*, Catherine Plaisant, Jennifer Golbeck*, Ben Shneiderman

Int. J. Hum. Comput. Interaction, 2013

human-computer interaction

Improving Healthcare with Interactive Visualization

Ben Shneiderman, Catherine Plaisant, Bradford W. Hesse*

IEEE Computer, 2013

visualization

Interactive Information Visualization to Explore and Query Electronic Health Records

Alexander Rind, Taowei David Wang, Wolfgang Aigner, Silvia Misch, Krist Wongsuphasawat, Catherine Plaisant, Ben Shneiderman

Foundations and Trends in Human-Computer Interaction, 2013

visualization, information visualization, bioinformatics

Temporal Event Sequence Simplification

Megan Monroe, Ronglan Lan, Hanseung Lee, Catherine Plaisant, Ben Shneiderman

IEEE Trans. Vis. Comput. Graph. [VAST], 2013

visualization, visual analytics

The challenges of specifying intervals and absences in temporal queries: a graphical language approach

Megan Monroe, Ronglan Lan, Juan Morales del Olmo*, Ben Shneiderman, Catherine Plaisant, Jeff Milstein*

2015 ACM SIGCHI Conference on Human Factors in Computing Systems, CHI '15, Paris, France, April 27 - May 2, 2015, 2015

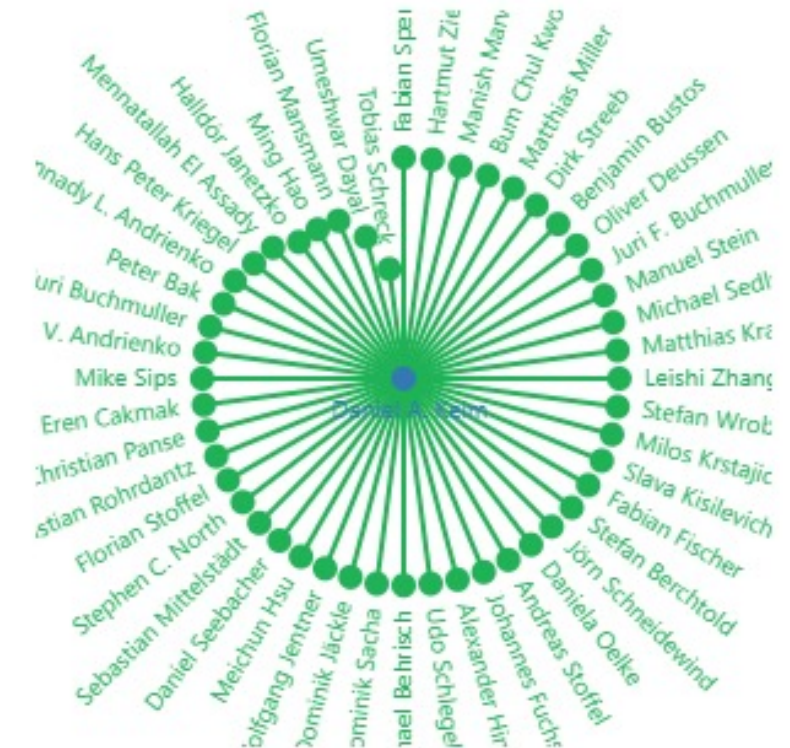
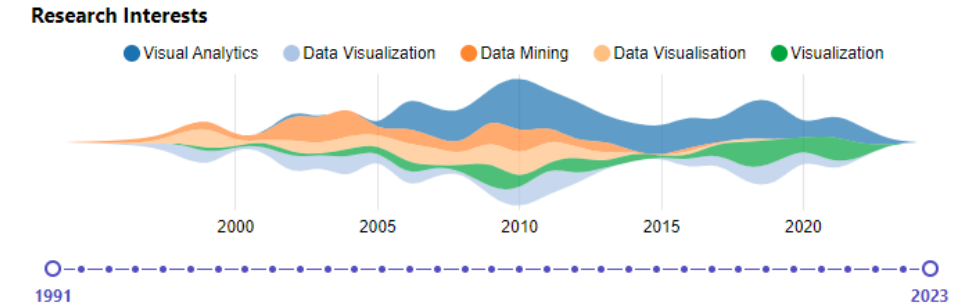
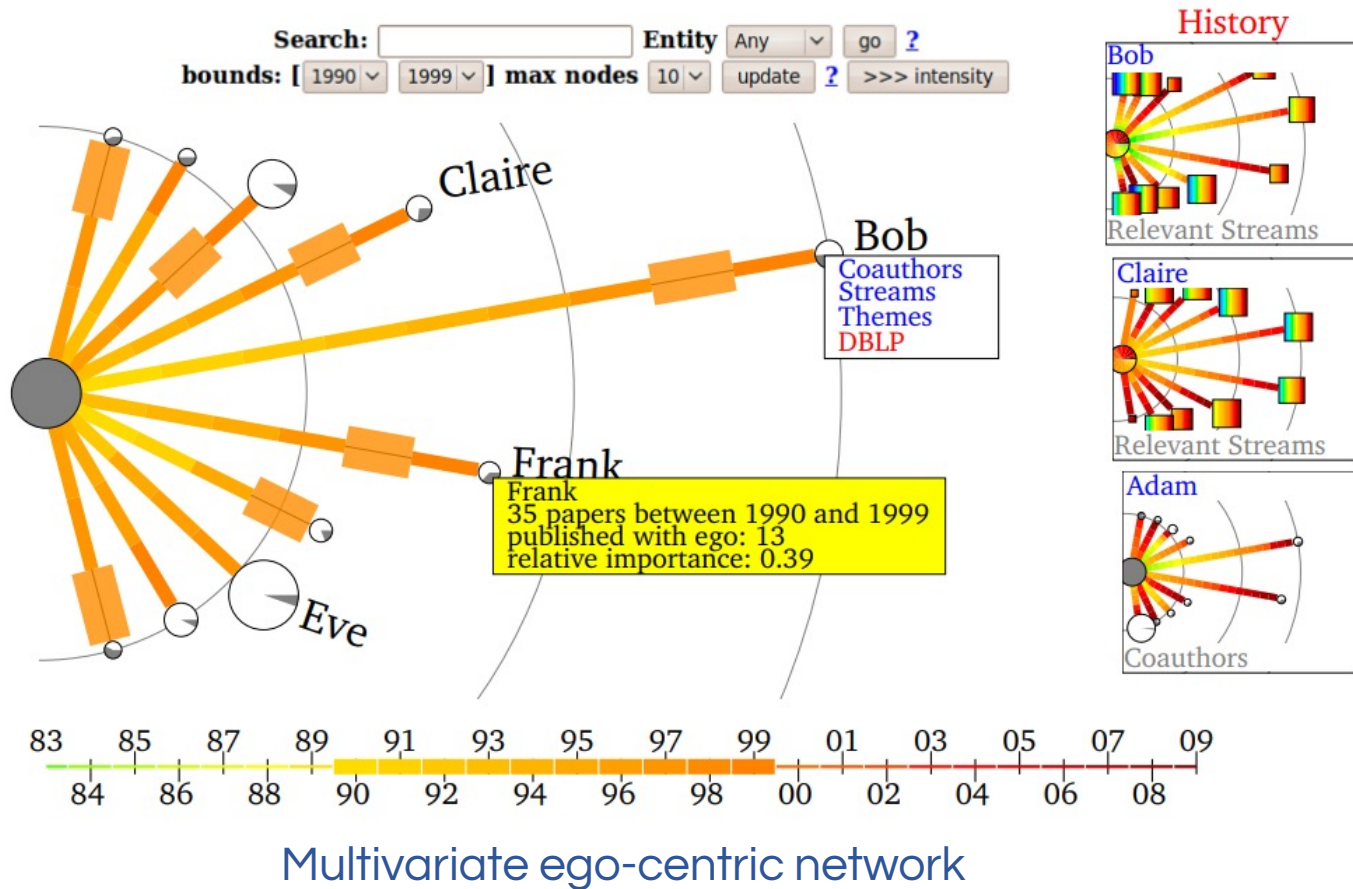
human-computer interaction

Twins: Novel User Interface Designs for Medication Reconciliation

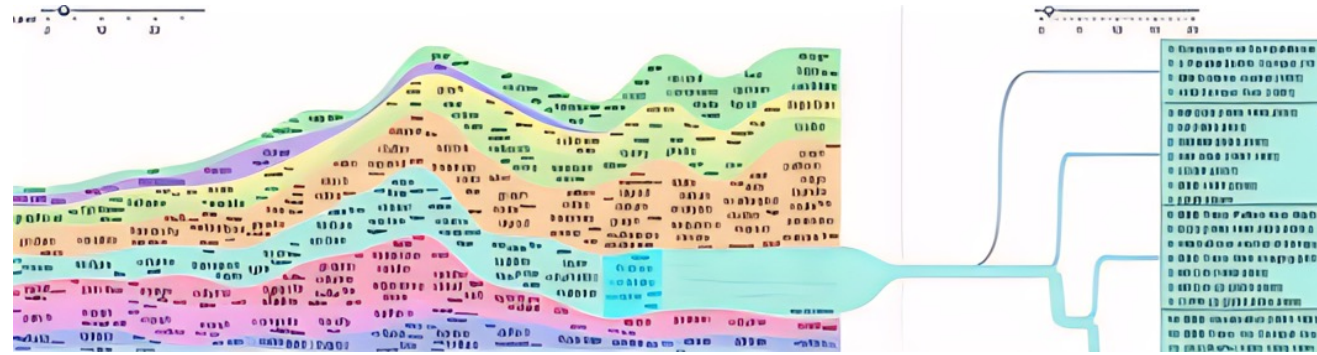
Catherine Plaisant, Tiffany Chao*, Johnny Wu*, Aaron Z.

Related Work: Collaboration network visualization

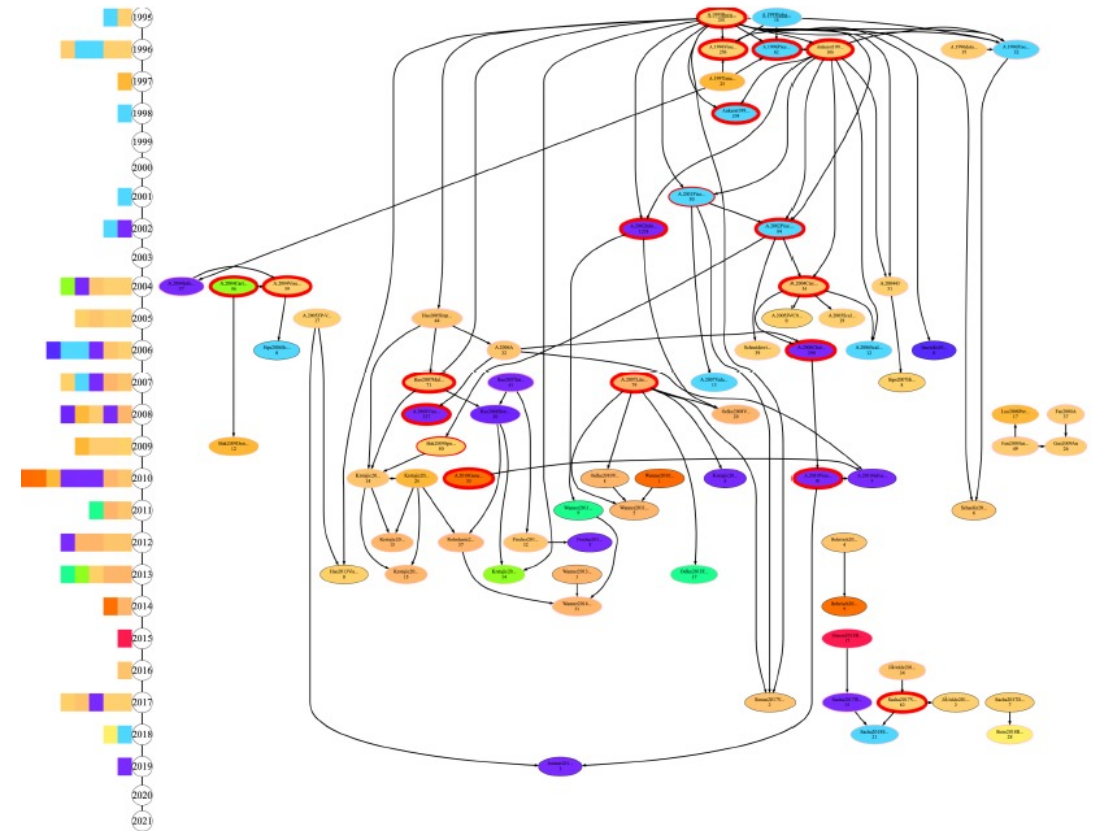
- Reveal **contextual information**, but few focuses on scholar-centric research evolution



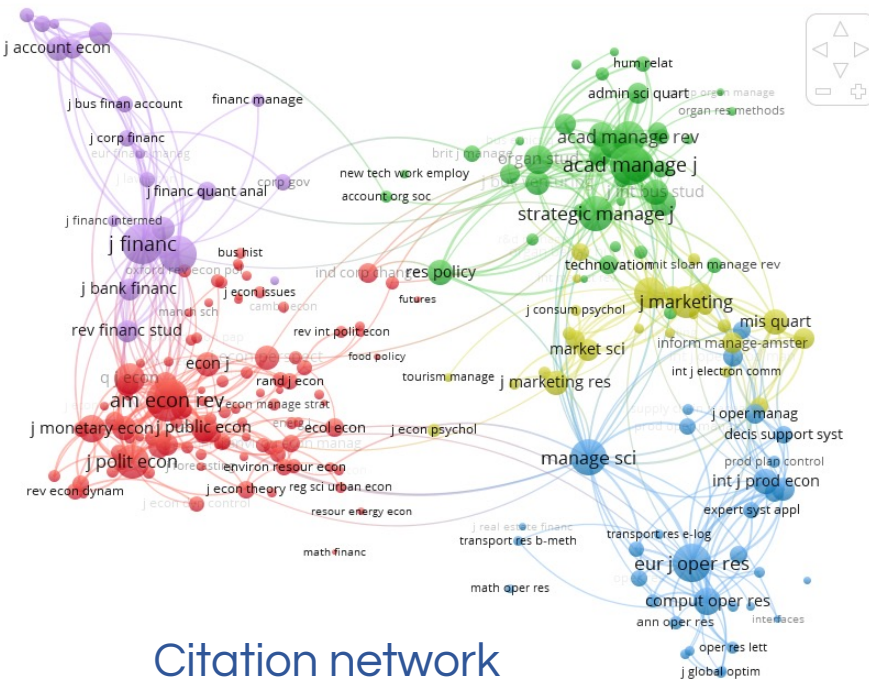
Related Work: Visual analysis of citation-based profile



Topic evolution



➤ GeneticFlow utilize curated self-citation data to illustrate the scientific research evolution of individual scholars



Citation network



Co-citation analysis

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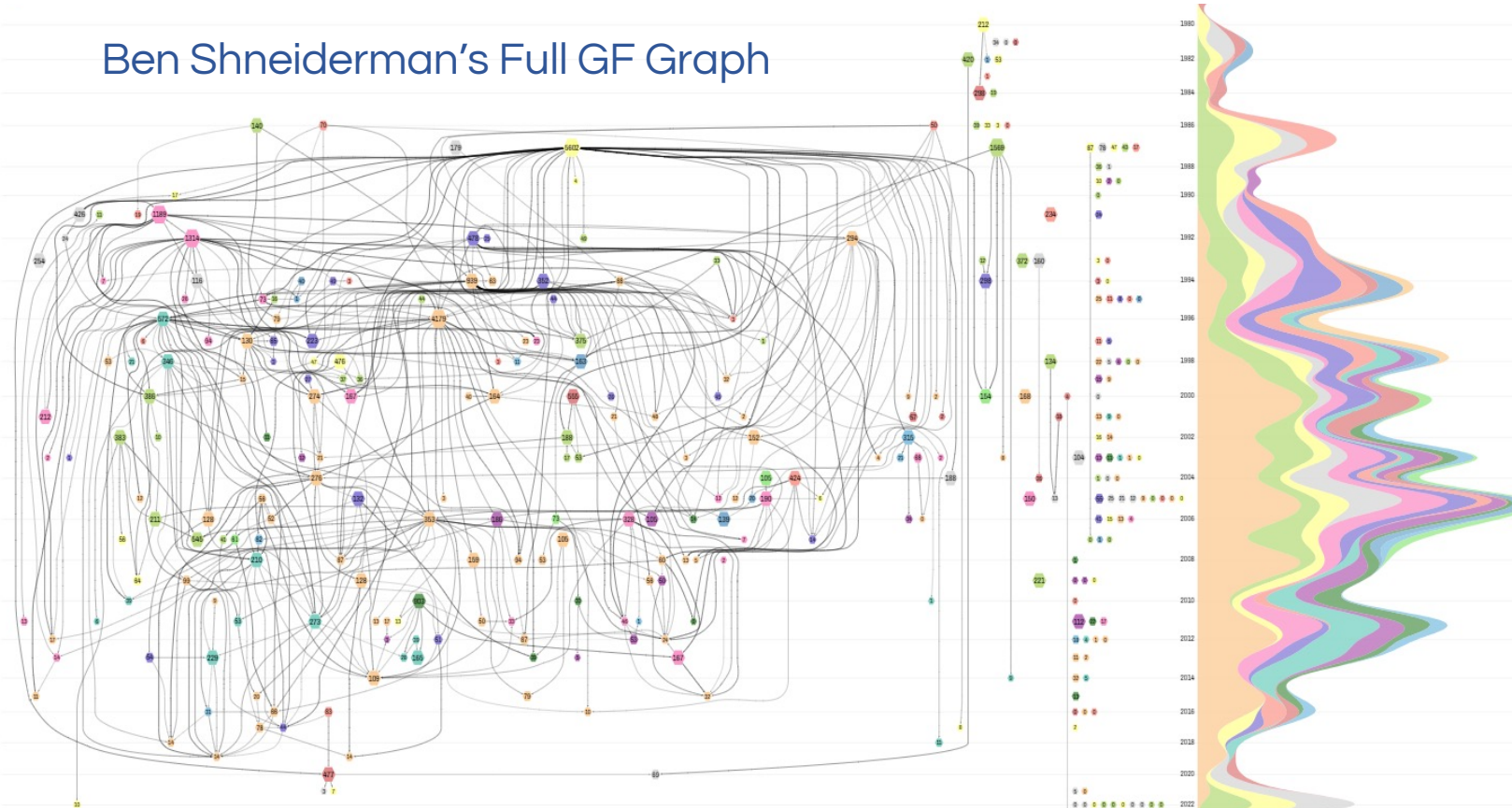
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Ben Shneiderman's Full GF Graph



data visualization analysis

interface user design

node treemaps visualization

data health visualization

human computer interaction

search web information

interaction display

support work

social community

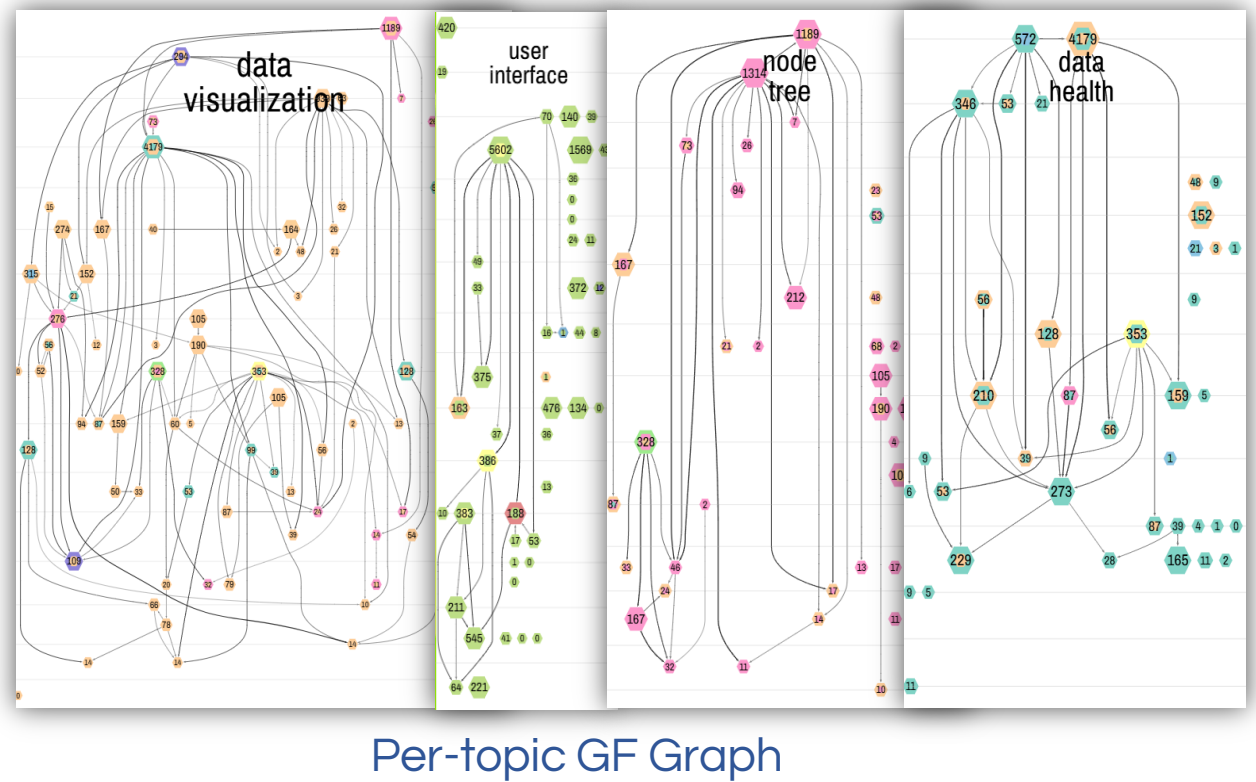
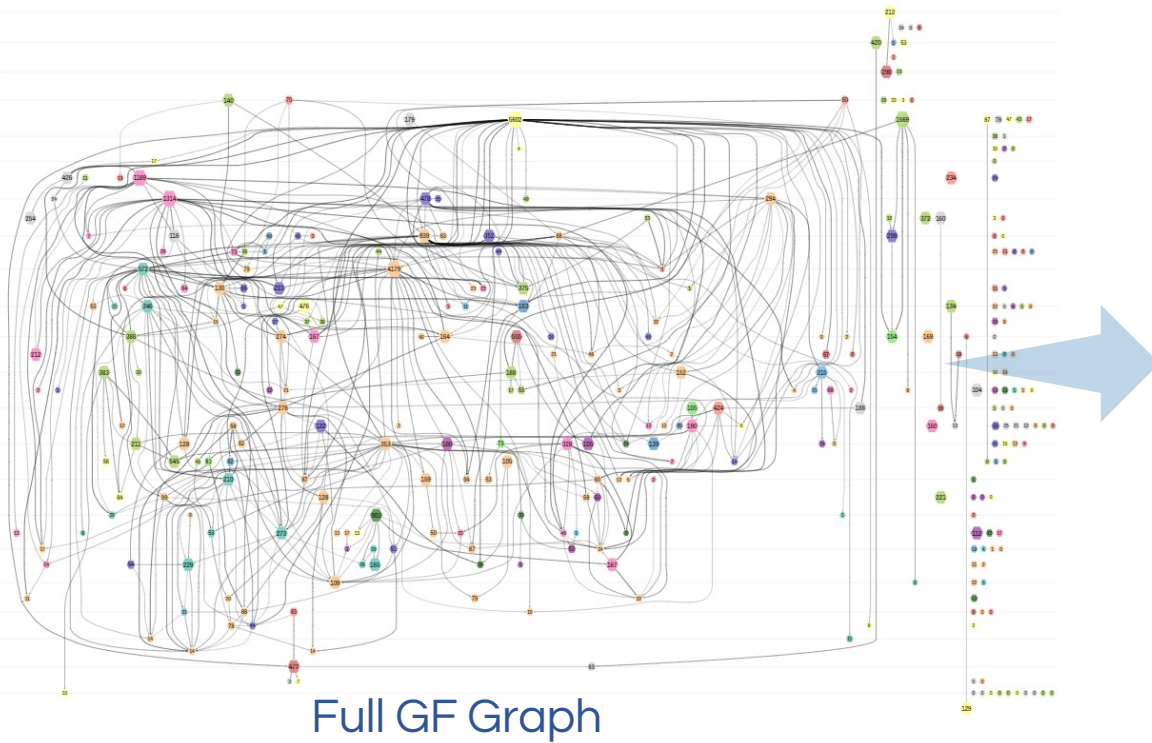
semantic visual

data visualization

social twitter

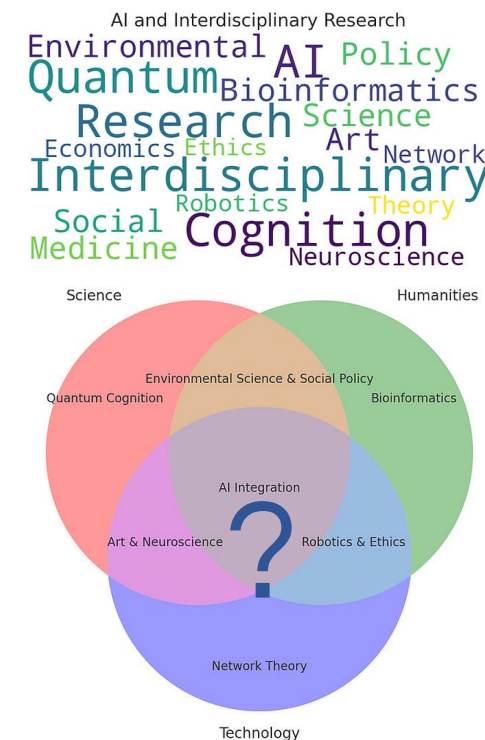
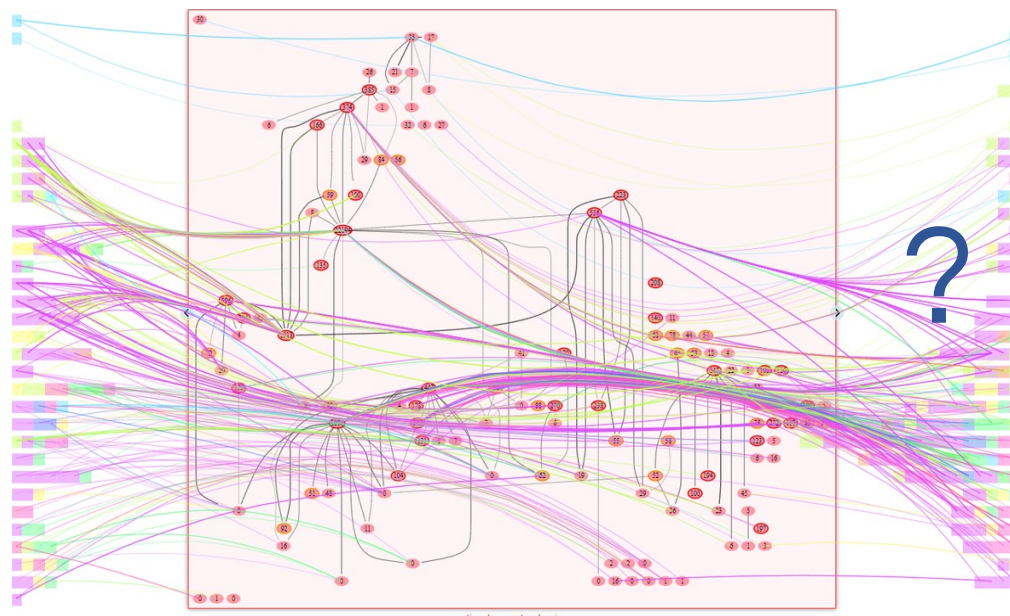
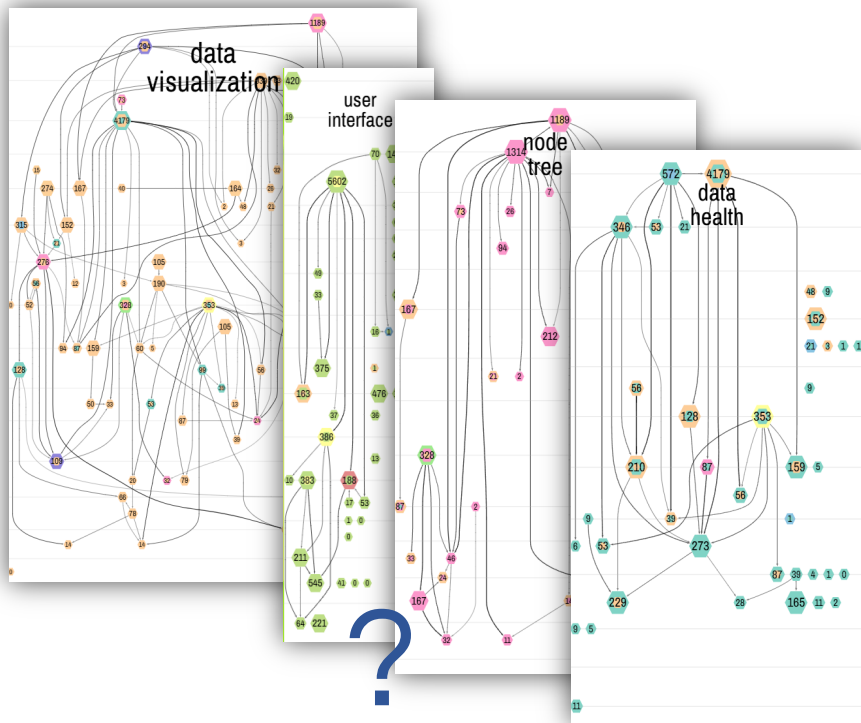
- Each research is assigned with one topic without considering **topic interaction**
- Fail to explicitly display **interdisciplinary activities** or **research transformations**
- Suffer from huge **visual clutter and edge crossings** for top scholars

Motivation



- Scholars' works on the same research topic are normally **more interconnected**
- Splitting a full GF graph into **multiple topic facets** and displaying them separately
 - Topic facets: well-established research topics
 - Multiple topic facets: engage in several distinct topics concurrently

Challenge



- How to visually **accommodate multiple topic facets** in a traditional 2D display space?
- How to visualize topic-based evolution while still **uncovering inter-topic interactions**?
- How to display **interdisciplinary works** of a scholar's research evolution?

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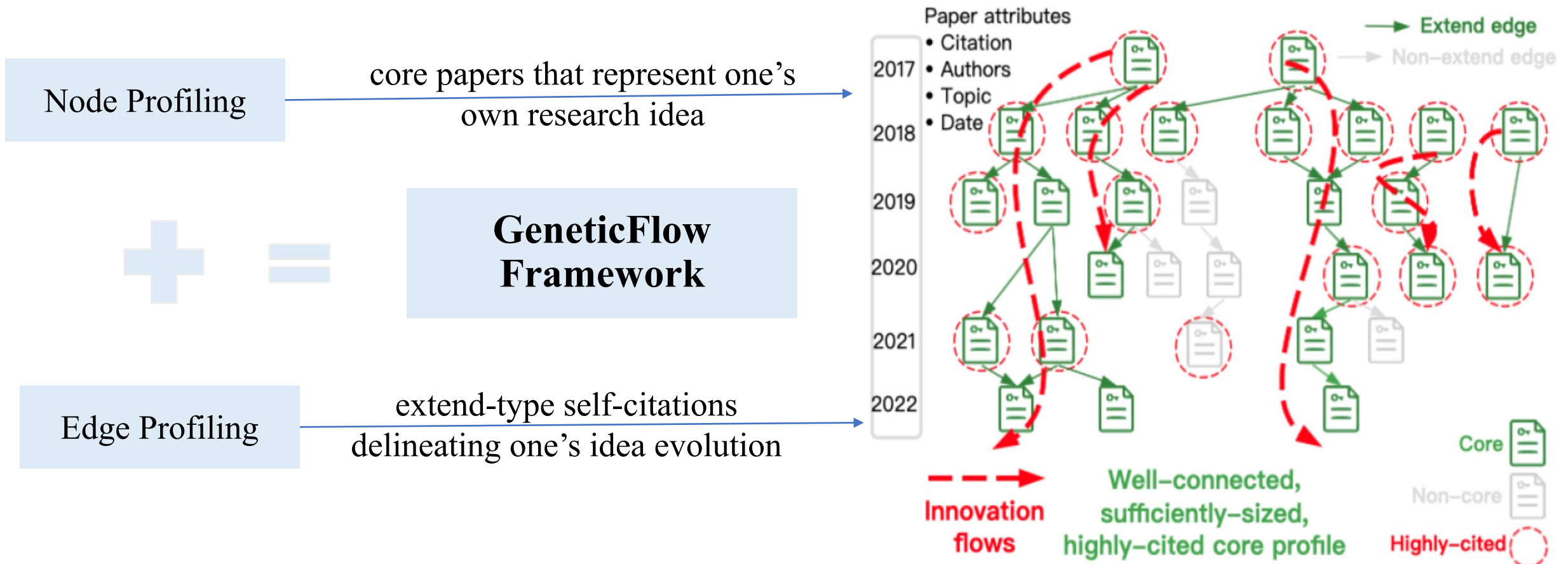
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GeneticFlow Graph Model

- GeneticFlow (GF) Graph Model: Explore the self-citation graph of a scholar which connects all publications into a structured context for interpreting scholar's impact.



GeneticFlow Graph Model

Node Profiling

- A paper is considered a core paper if the scholar is
 - first author
 - supervisor of the first author

$$p_{\text{cont}}(a_k) = \begin{cases} 1, & \text{if } k = 1 \\ p_{AA}(a_k, a_1, t), & \text{if } k \neq 1 \end{cases}$$

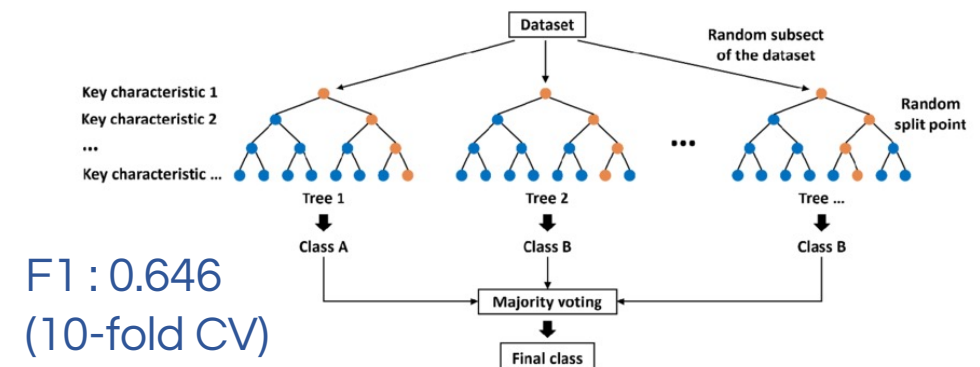


Edge Profiling

- Four categories of 20 features

metadata of cited and citing papers	features extracted from their citation networks
temporal correlation measures	content and lexical patterns from the citation context

- Extra-Tree Model as classifier



User & Task Characterization



Researchers

Using data-driven analysis to track academic progress, aiding future topics, collaborations, and research trends.



Academic administrators

Handling hiring, awards, funding, or field agendas, benefiting from GeneticPrism's data-enhanced peer review.

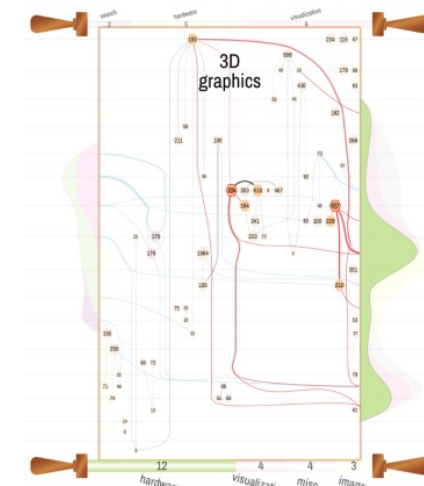
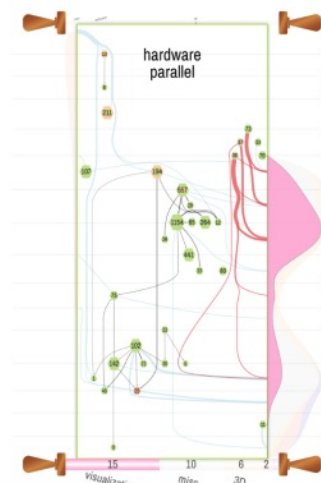
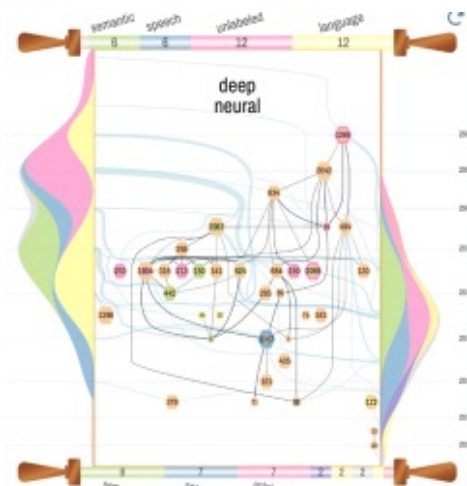
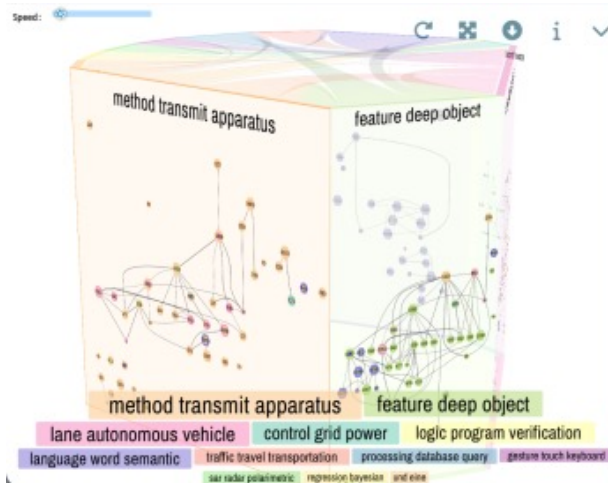
Task

T1: temporal and topical overview of a scholar's research.

T2: detailed analysis of a scholar's research evolution on a single research topic

T3: discover influence and interaction patterns among multiple research topics of a scholar.

T4: identifying key interdisciplinary papers and their influence patterns



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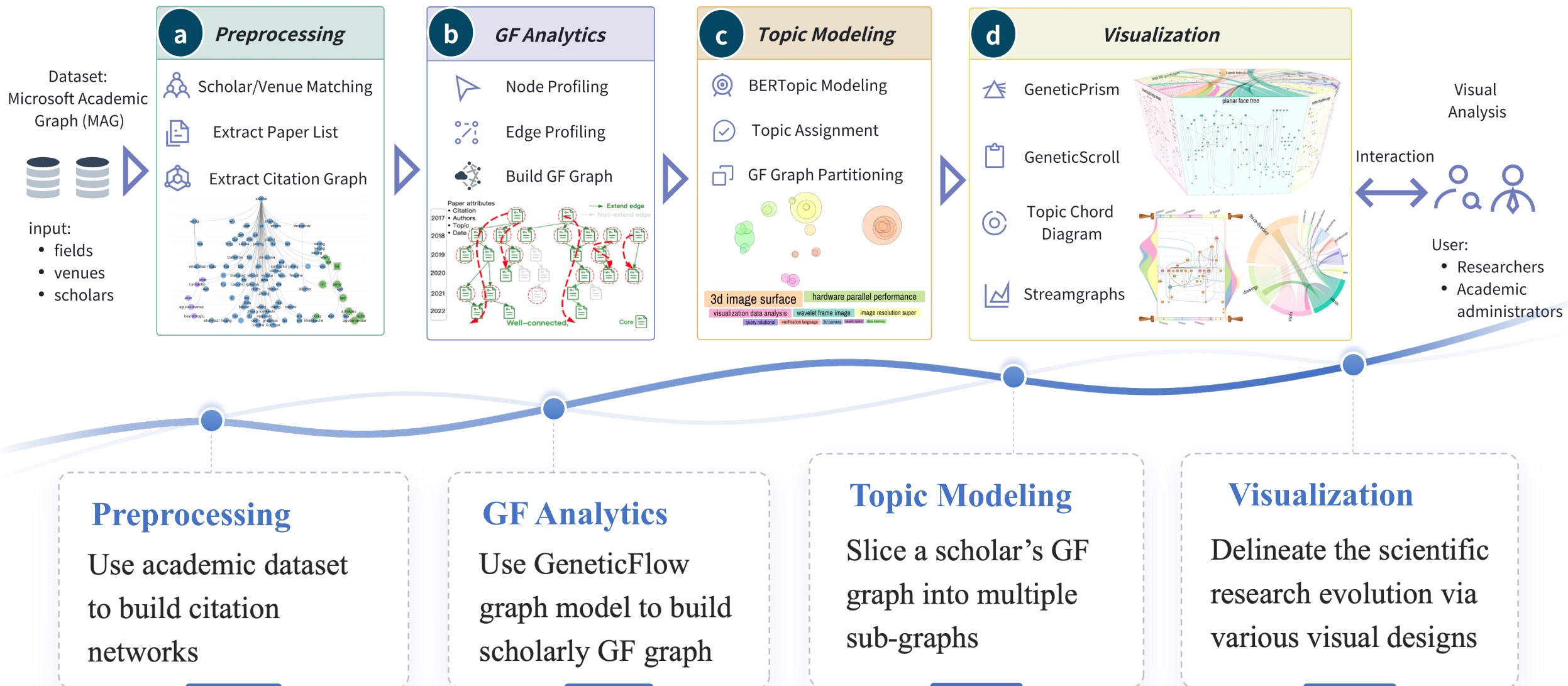
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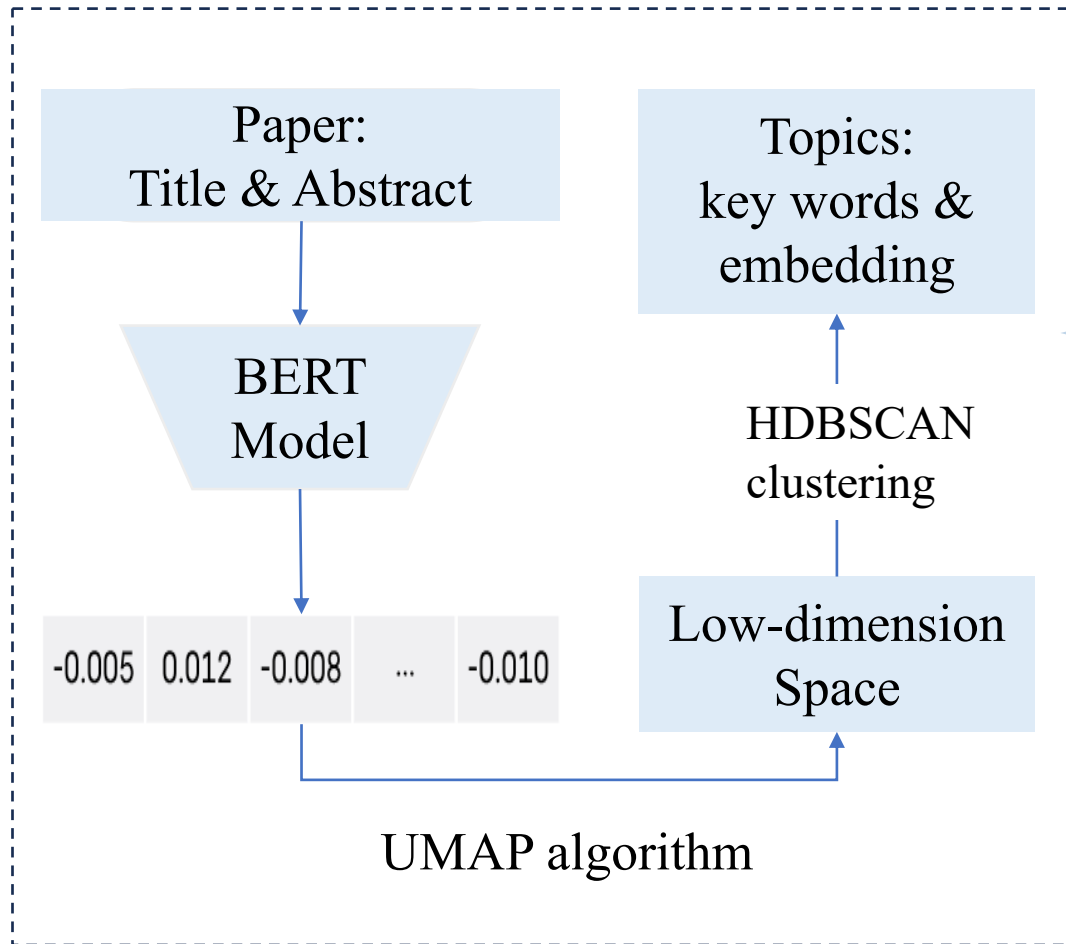
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Pipeline

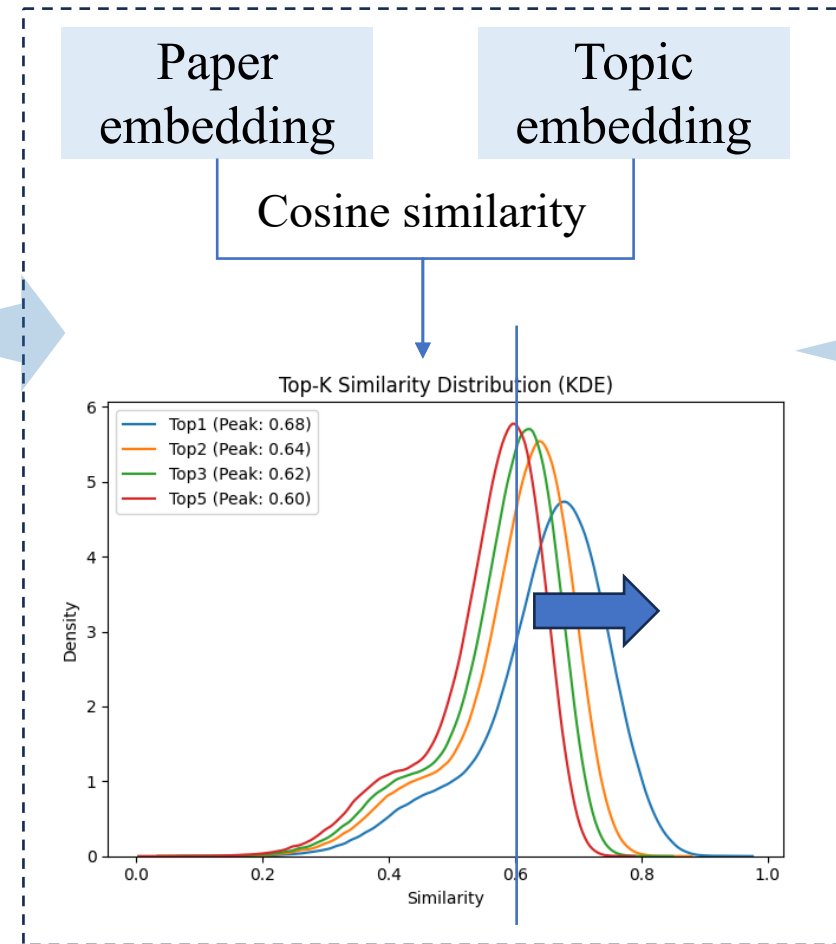


Topic Modeling

Bertopic Modeling

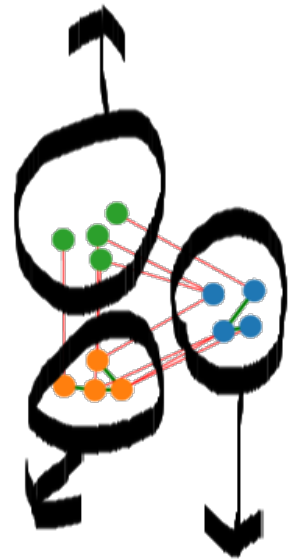


Topic Assignment

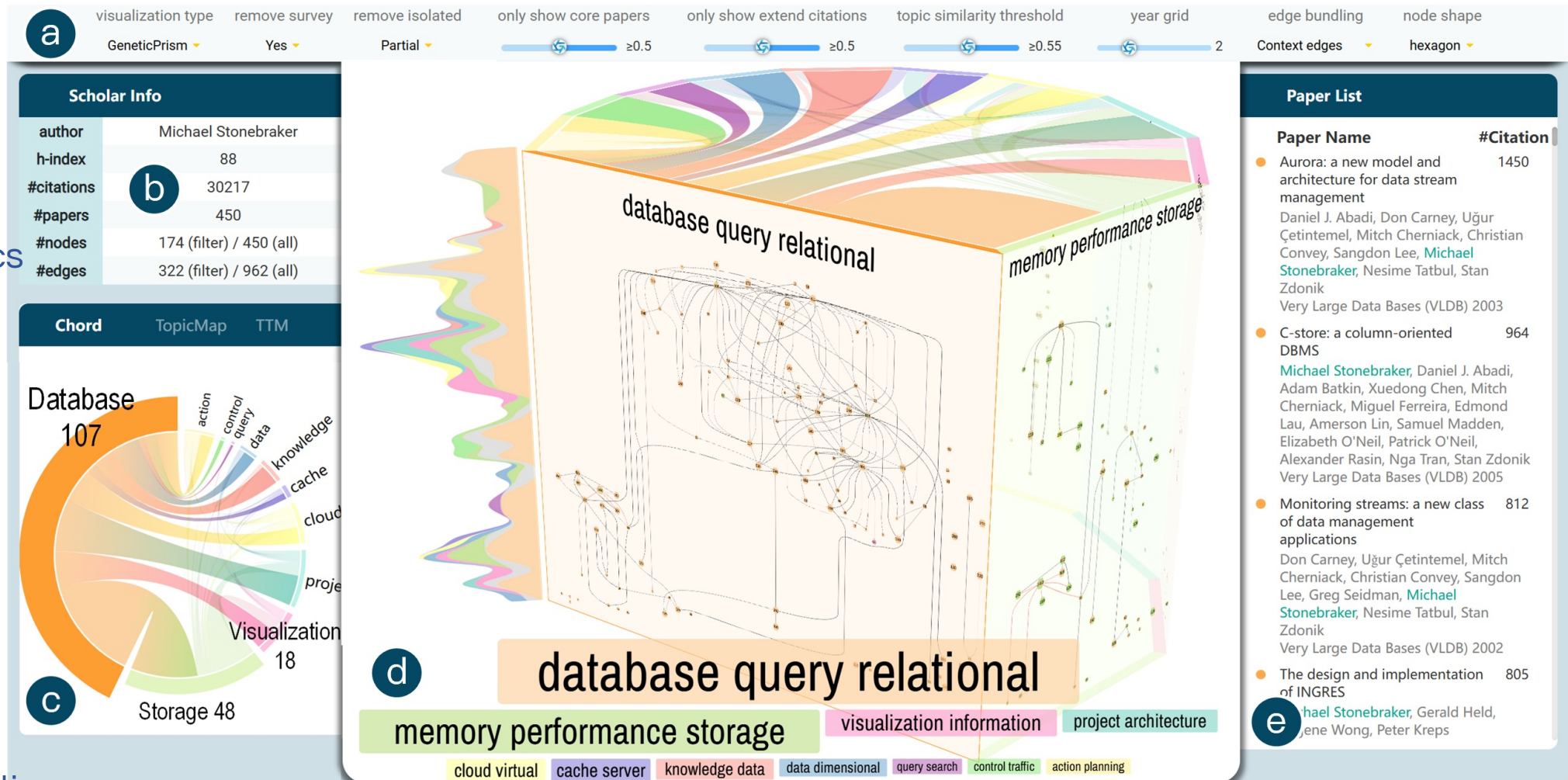


Assigned to top 3 topics for which the similarity exceeds a threshold

GF graph partitioning



Visualization controller

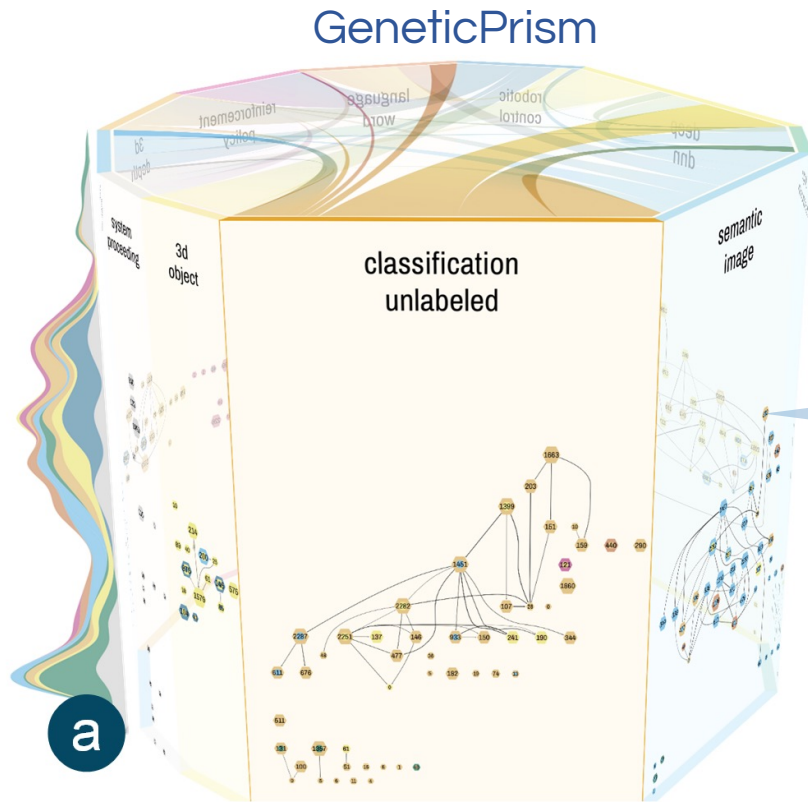


Demographics panel

Topic interaction panel

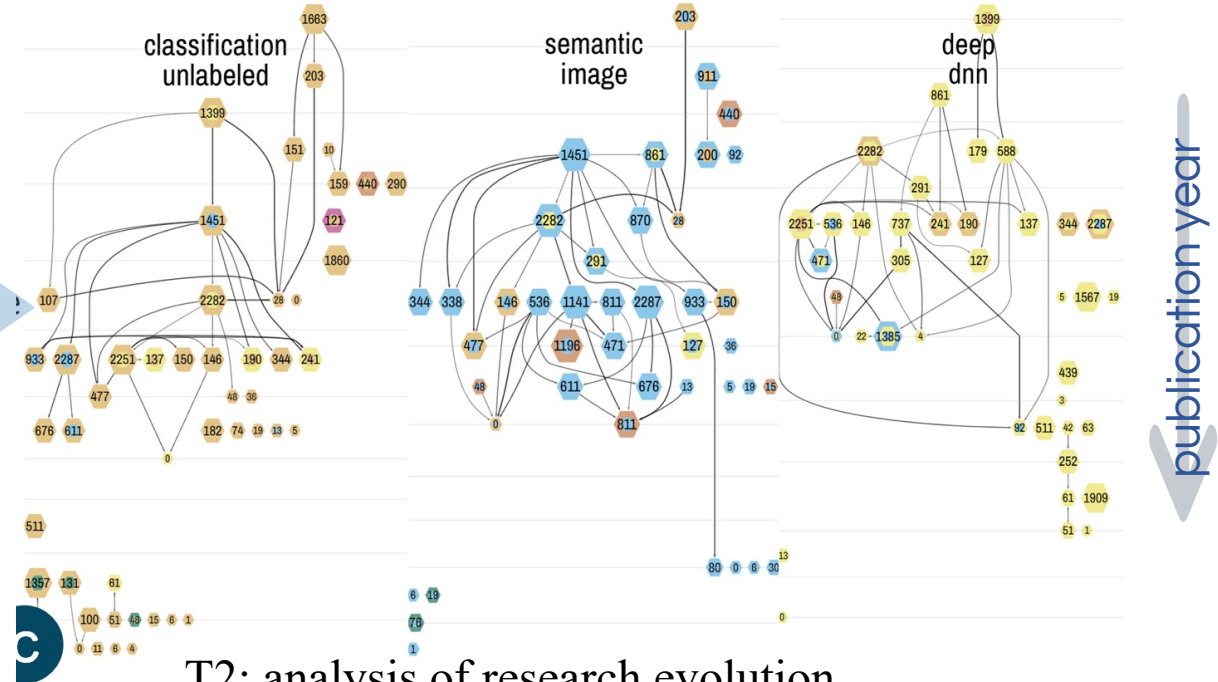
Main panel: visualize GF graphs (GeneticPrism/GeneticScroll)

Paper list panel



T1: temporal and topical overview

Side view: per-topic GF subgraphs

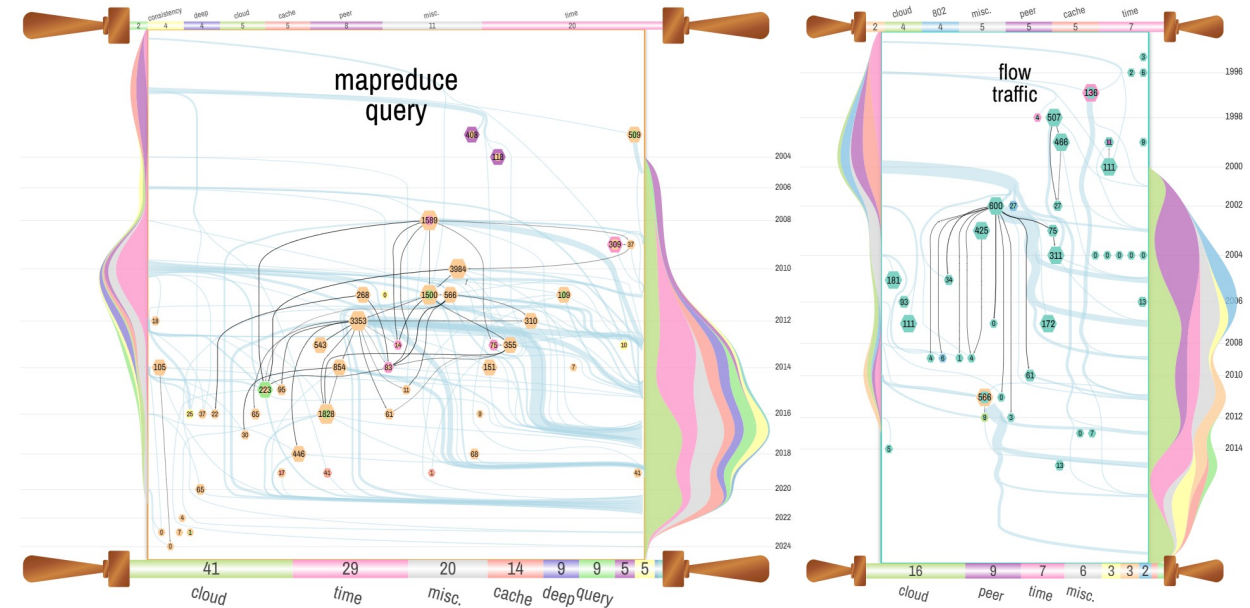
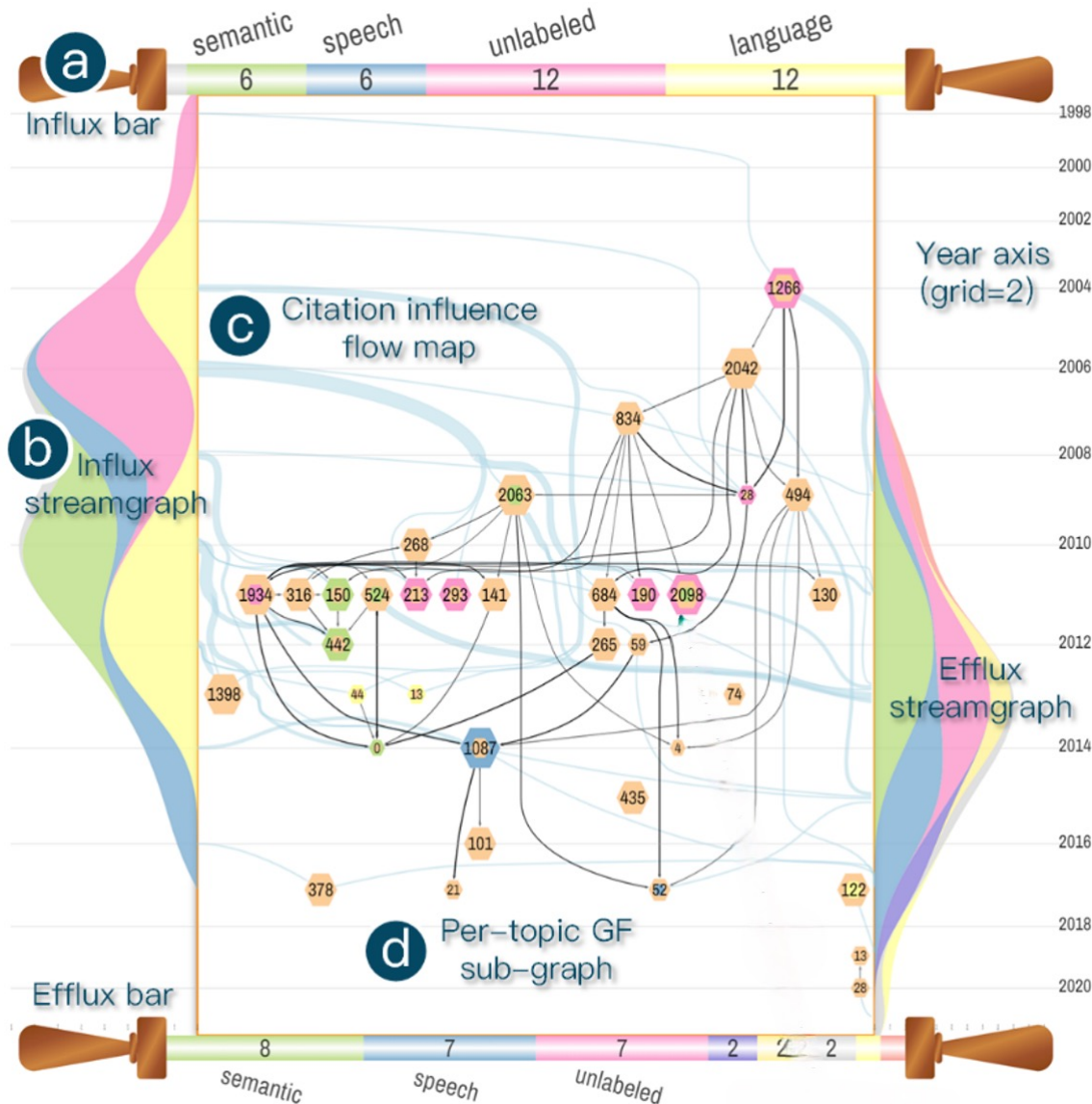


T2: analysis of research evolution on a single research topic

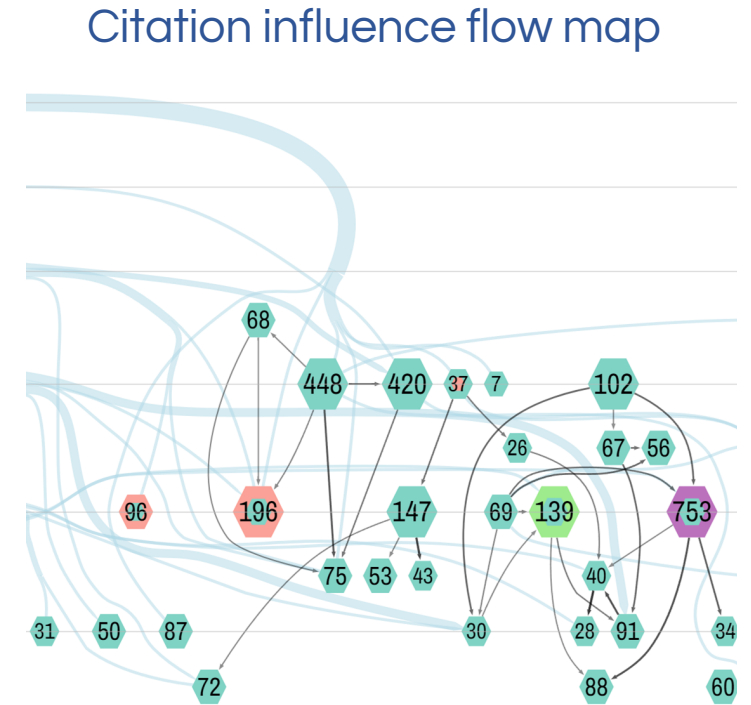
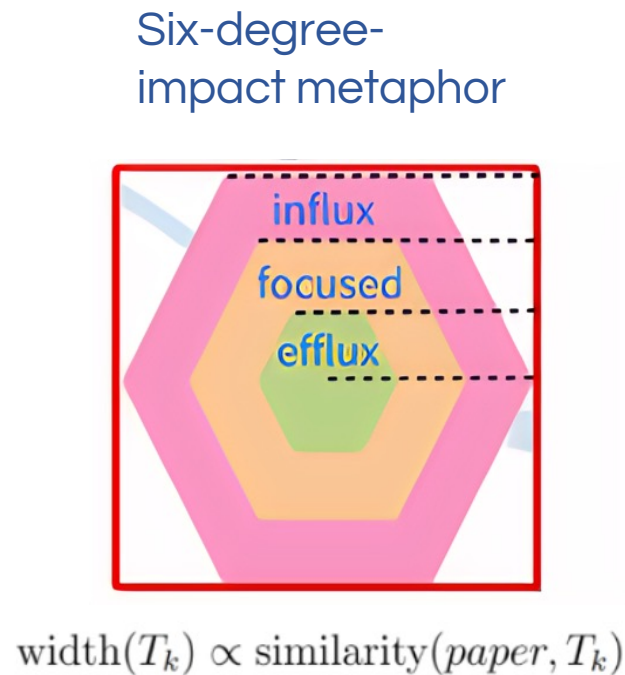
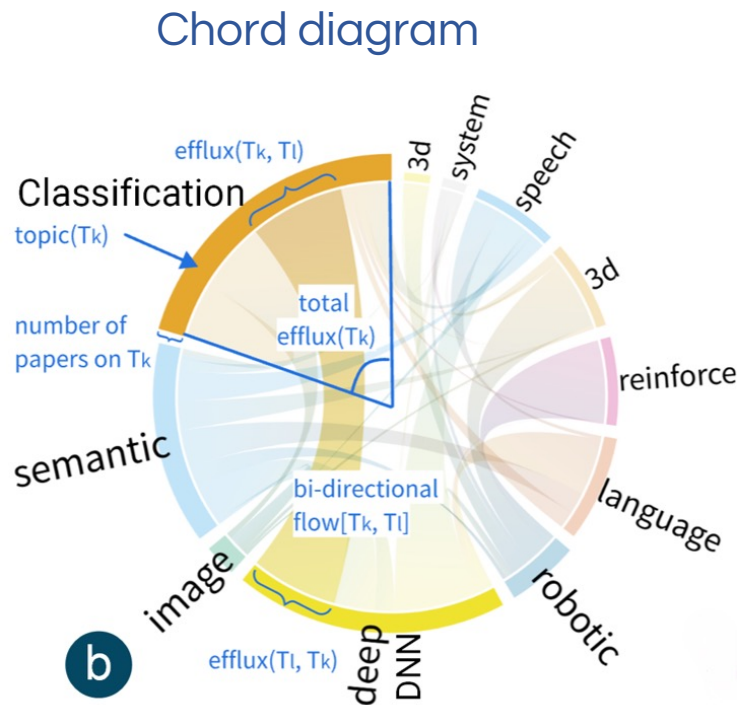
- Accommodate multiple topic facets in a 3D prism to display subgraph simultaneously
- Mimic a **cultural monument**, symbolizing the multifaceted and impactful scientific research contributions

Visualization: GeneticScroll

<https://genetic-flow.com>



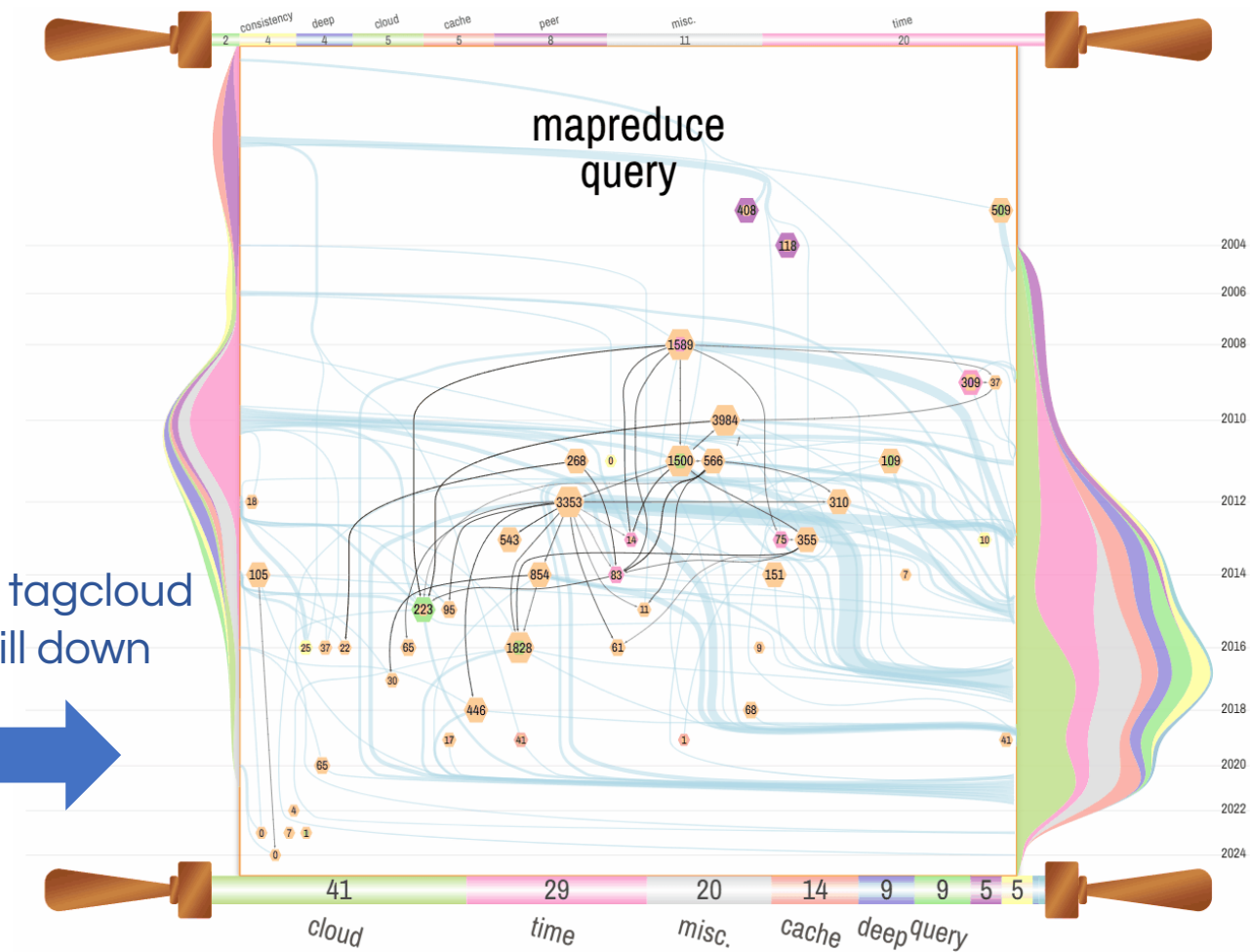
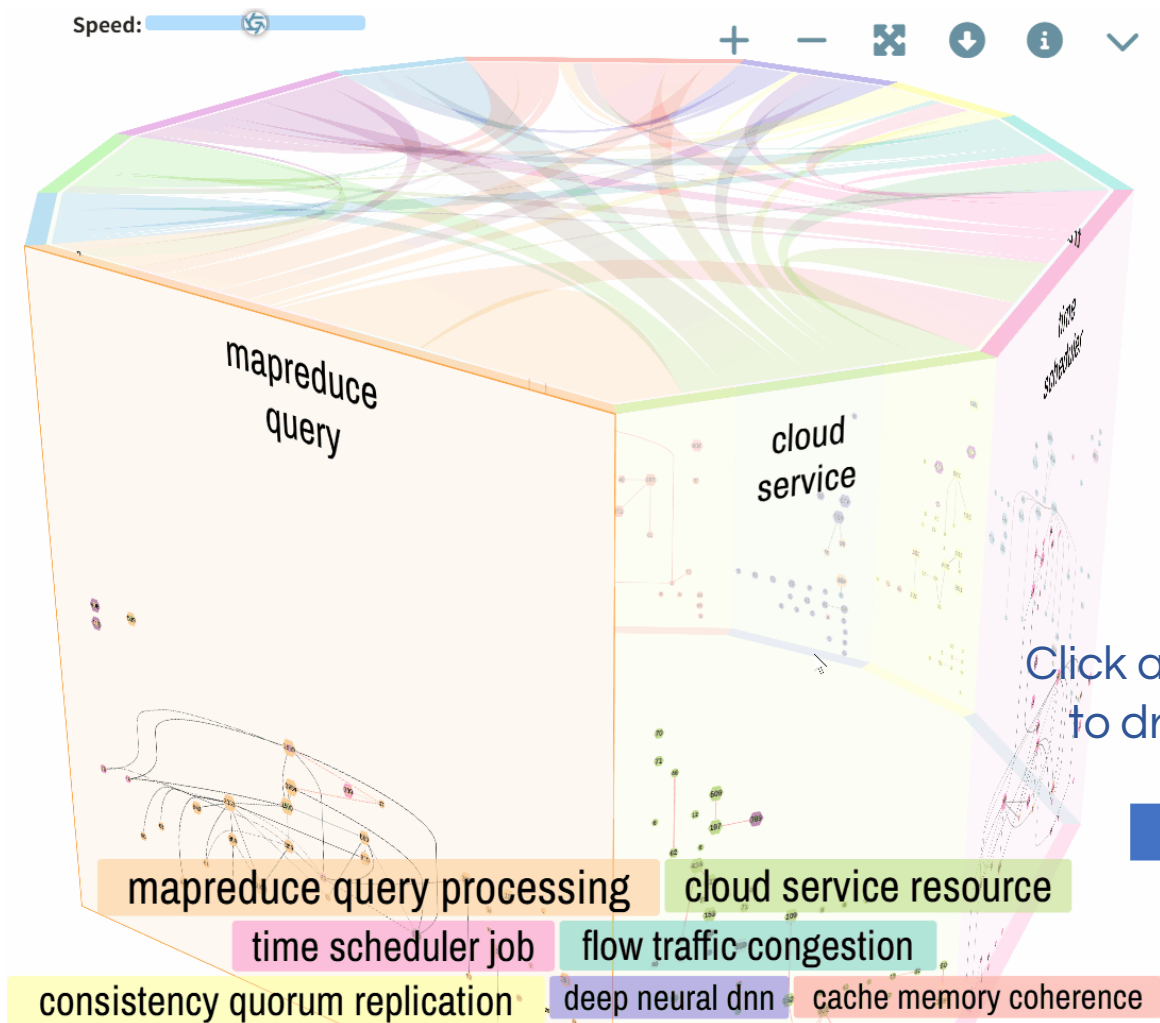
- Resembles a **traditional scroll painting**, highlighting the detailed and unfolding narrative of a scholar's research on a specific topic and its outreach



- Topic chord diagram: illustrates the citation influence graph among all derived research topics of a scholar (T_1)
- Layered six-degree-impact metaphor: the paper with overlapping topics, i.e., the interdisciplinary work, can be visually highlighted (T_4)
- Citation influence flow map reveals detailed topic-level interaction (T_3)

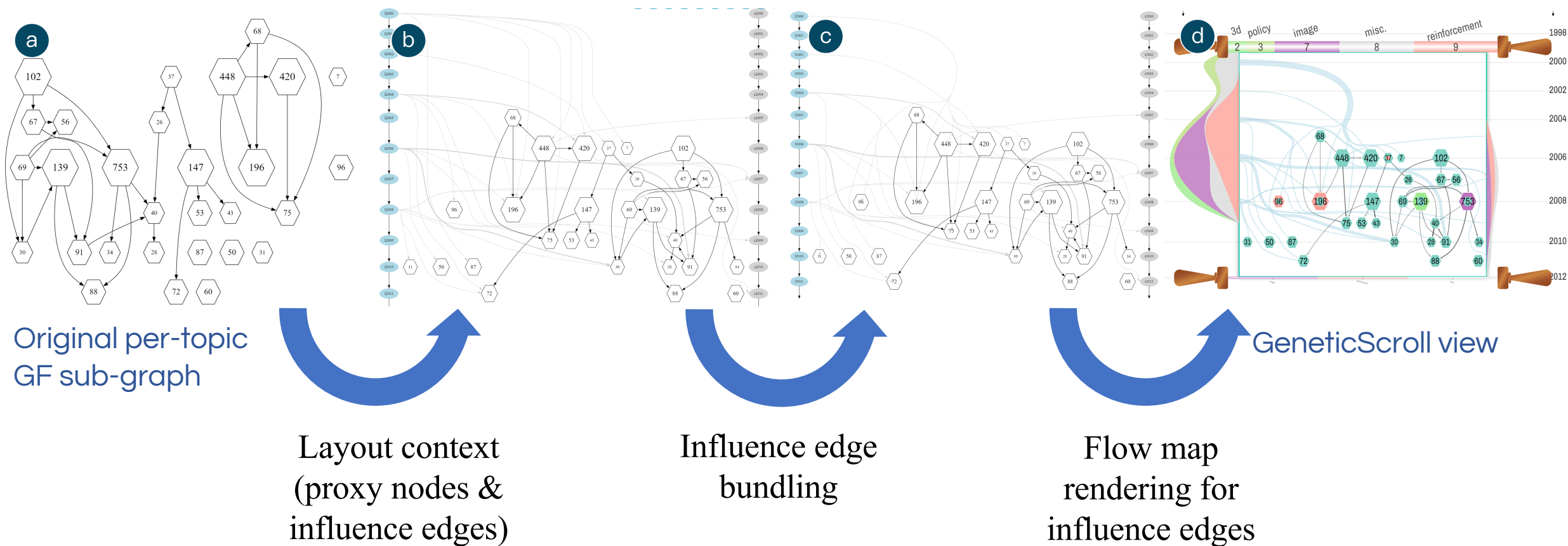
Interaction

<https://genetic-flow.com>



Flow-integrated Hierarchical Layout (FiHL) for GeneticScroll

- Integrate context as flow map into the original hierarchical layout

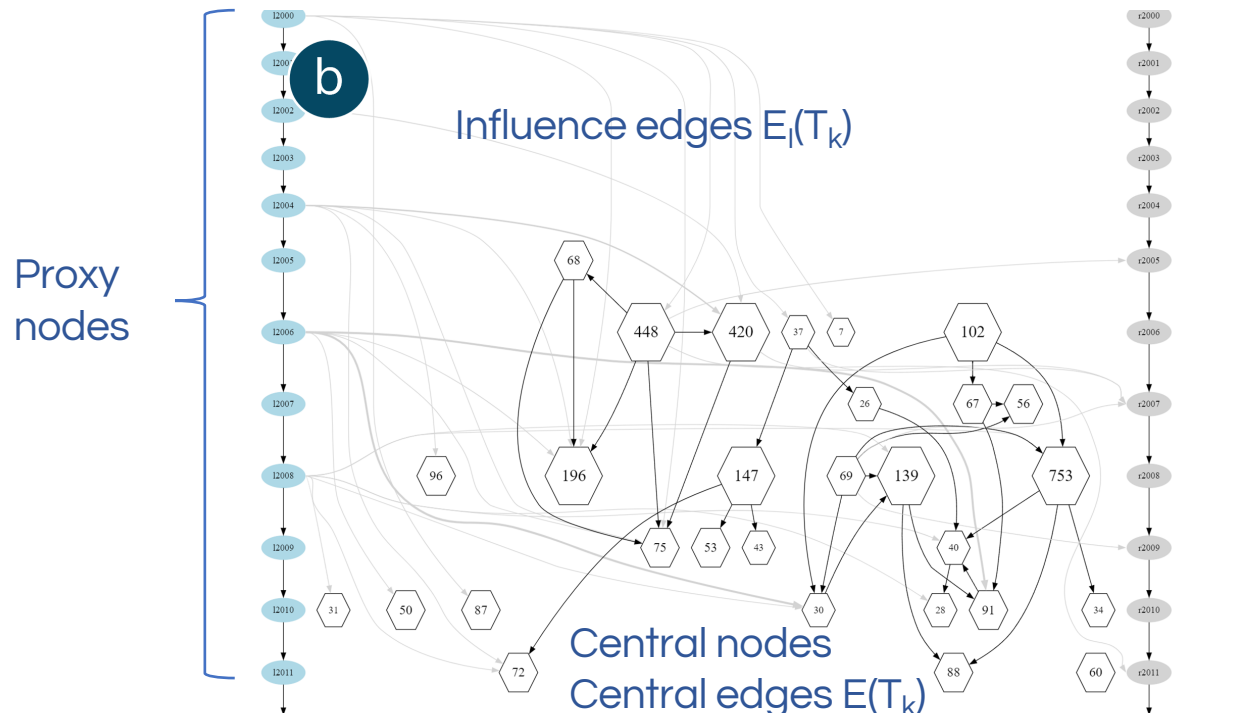


FiHL Algorithm: layout context

<https://genetic-flow.com>

- Add proxy nodes and influence edges
- Optimize weighted edge crossing

$$E_I(T_k) = \{e_{pj} \mid e_{ij} \in E, v_i \notin V(T_k) \wedge v_j \in V(T_k) \rightarrow v_p = v_l(y(v_i))\} \cup \{e_{ip} \mid e_{ij} \in E, v_i \in V(T_k) \wedge v_j \notin V(T_k) \rightarrow v_p = v_r(y(v_j))\}$$

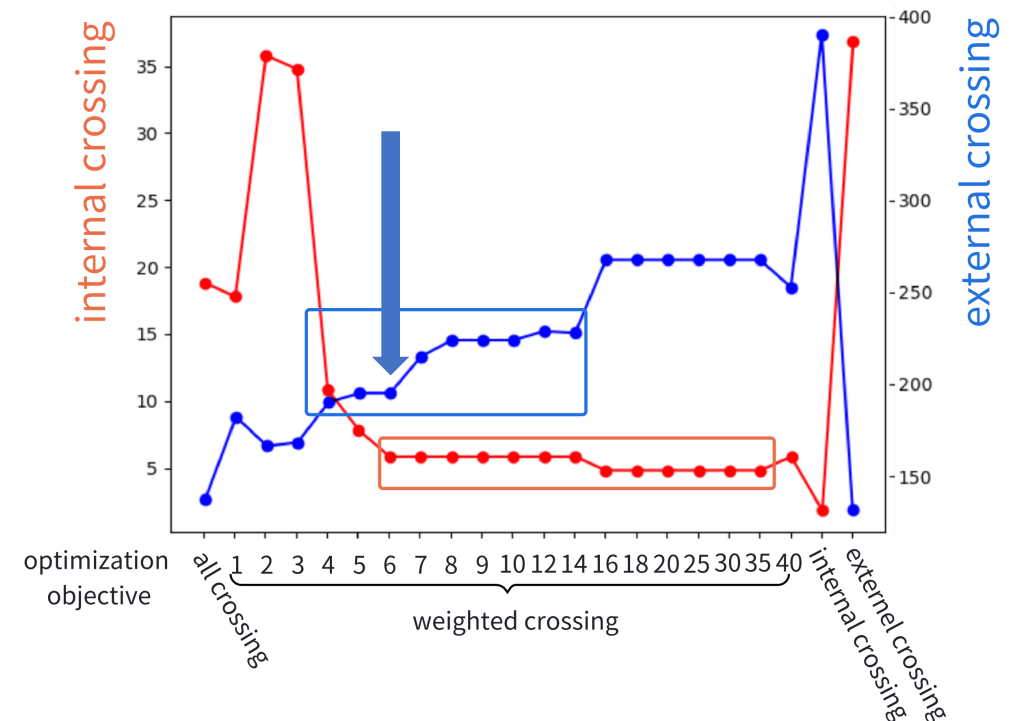


$$\text{cost}_{\text{all}}(e) = 1$$

$$\text{cost}_{\text{internal}}(e) = \begin{cases} 1 & \text{if } e \in E(T_k) \\ 0 & \text{if } e \in E_I(T_k) \end{cases}$$

$$\text{cost}_{\text{external}}(e) = \begin{cases} 0 & \text{if } e \in E(T_k) \\ 1 & \text{if } e \in E_I(T_k) \end{cases}$$

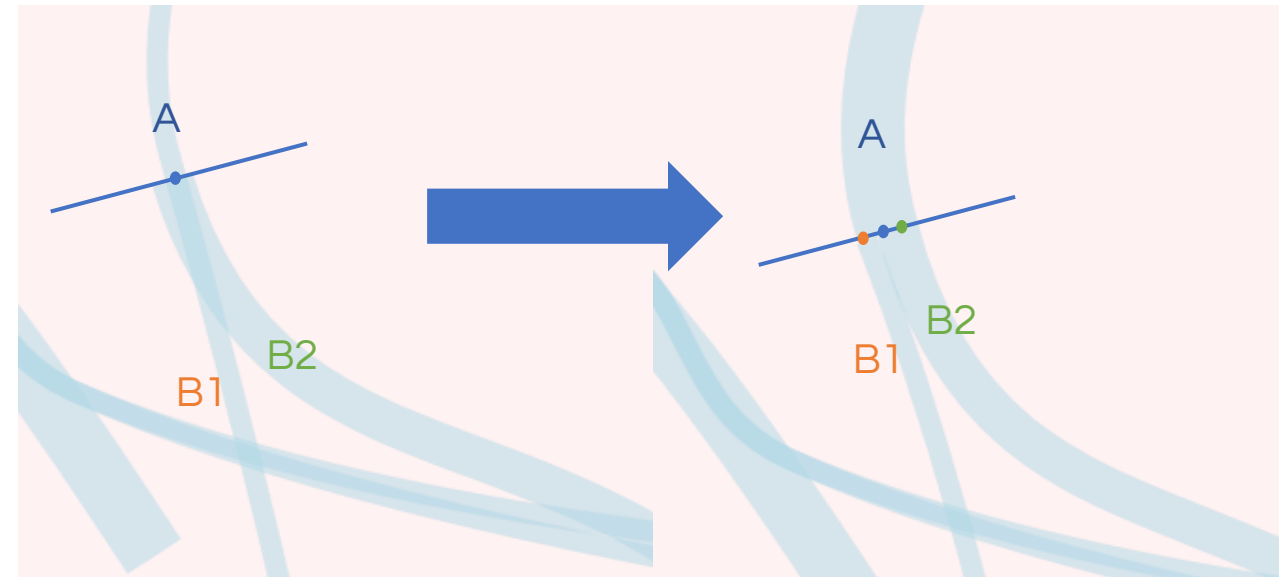
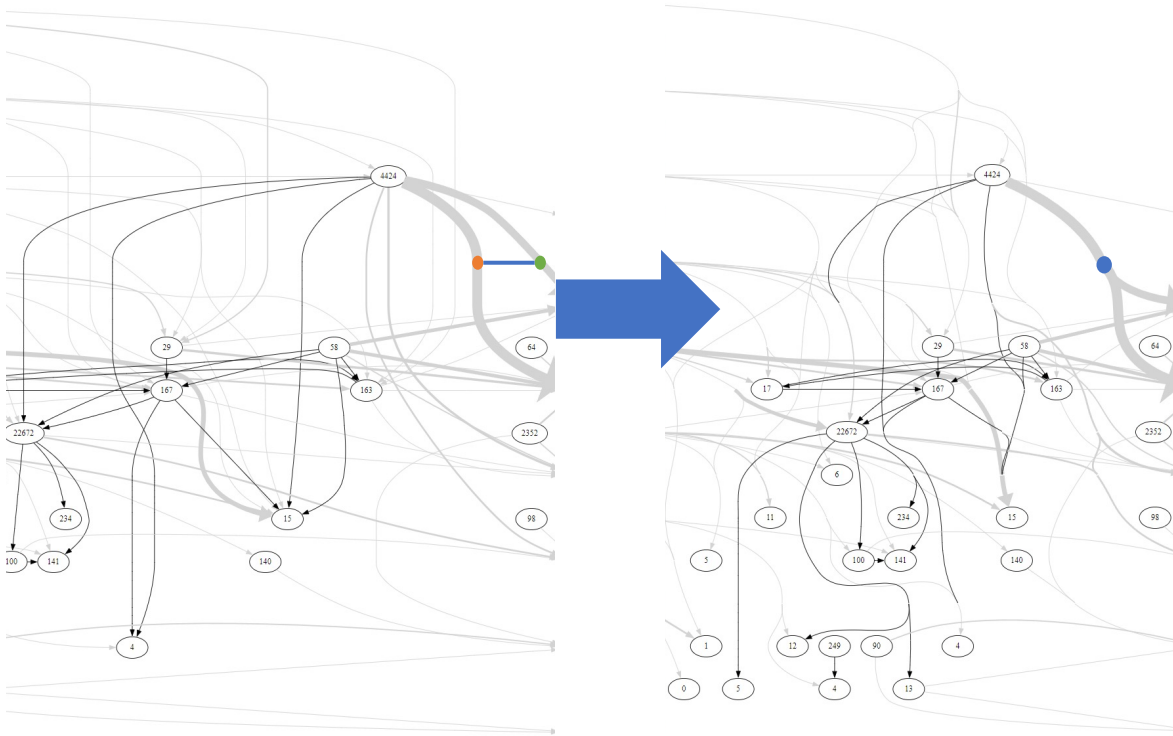
$$\text{cost}_{\text{weighted}}(e) = \begin{cases} \alpha & \text{if } e \in E(T_k) \\ \text{weight}(e) & \text{if } e \in E_I(T_k) \end{cases}$$



FiHL Algorithm: edge bundling as flow map

<https://genetic-flow.com>

- Edge bundling by merging adjacent dummy nodes (only influence edges)
- Adjust control points of Bézier curves to generate a Sankey-style diagram



A: M67.03,-1018.32C114.96,-1018.32 76.24,-950.74 **114.03,-921.26**
B1: M**114.03,-919.26**C147.7,-893.01 101.72,-864.09 114.03,-823.21
B2: M**114.03,-919.26**C123.21,-790.76 131.66,-754.16 135.64,-736.2

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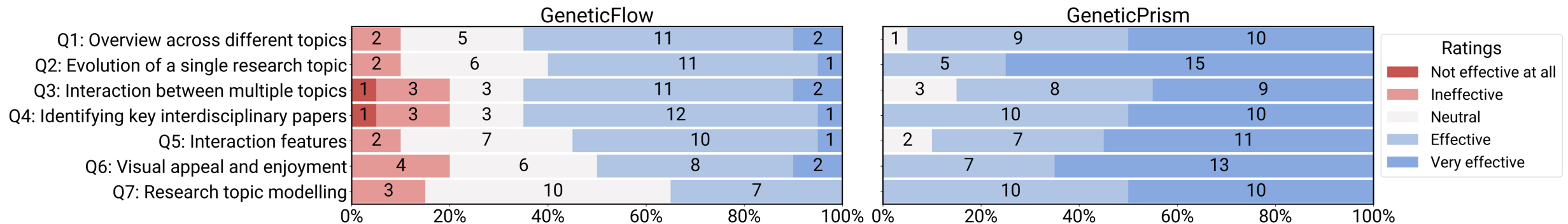
Evaluation

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User Evaluation

- In-lab user study using questionnaire on 20 participants (16 males, and 4 females) from diverse academic fields (databases, ML, VA, HCI). Compare GeneticPrism with the previous system GeneticFlow



significant improvements in task performance, interaction, topic modeling, and user engagement

- Open-ended questions: participants praised the
 - **3D prism** for effectively separating research topics
 - **the scroll** for analyzing inter-topic relationships and citation patterns
 - **the hexagonal node** design for encoding cross-disciplinary influences
 - **the multi-topic modeling** for capturing research complexity

Dataset & Preprocessing

➤ Academic Data Source:

➤ MAG (237 million papers, 240 million authors, and 1.63 billion citations)

➤ Preprocessing (eg. Top scholar in a subfield)

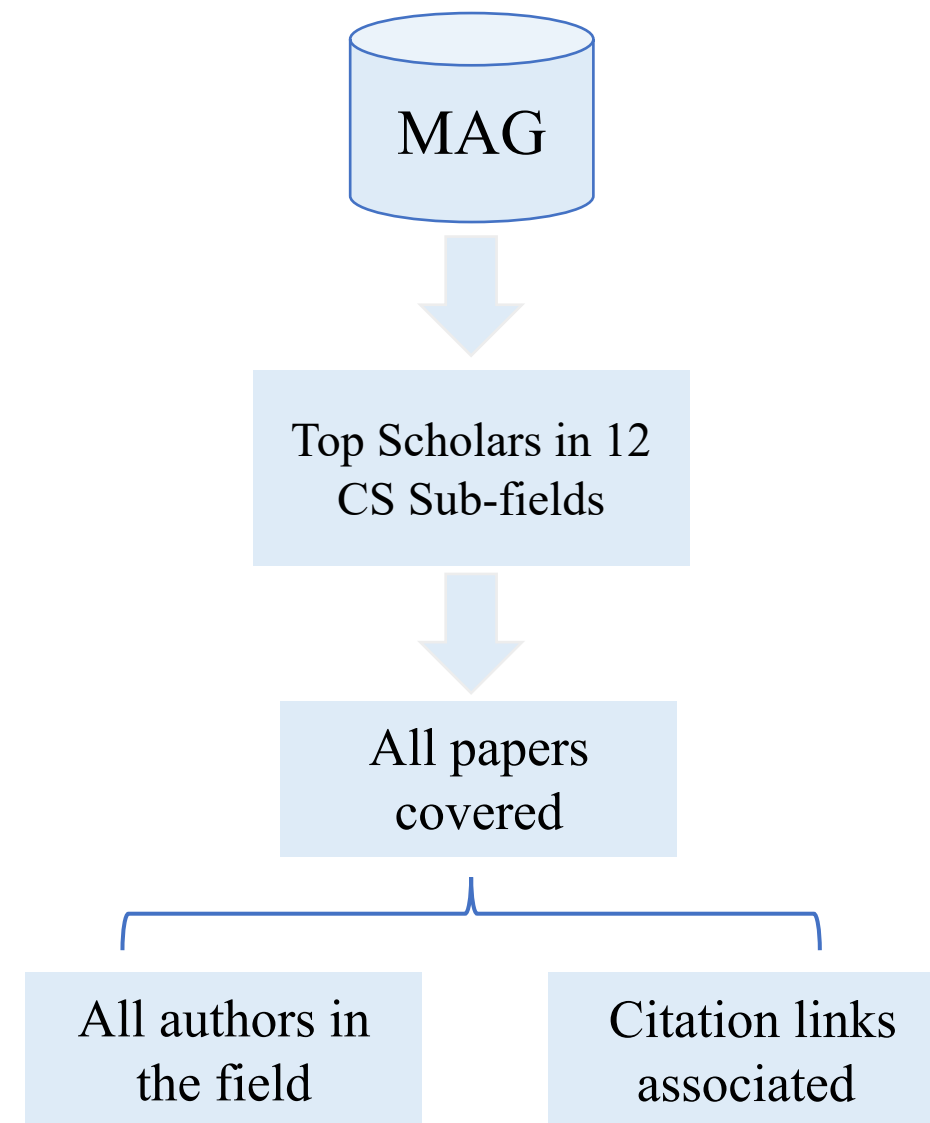
➤ Define subfield venues and topics

➤ Rank scholar by h-index and select top scholars

➤ Extract all papers and citations

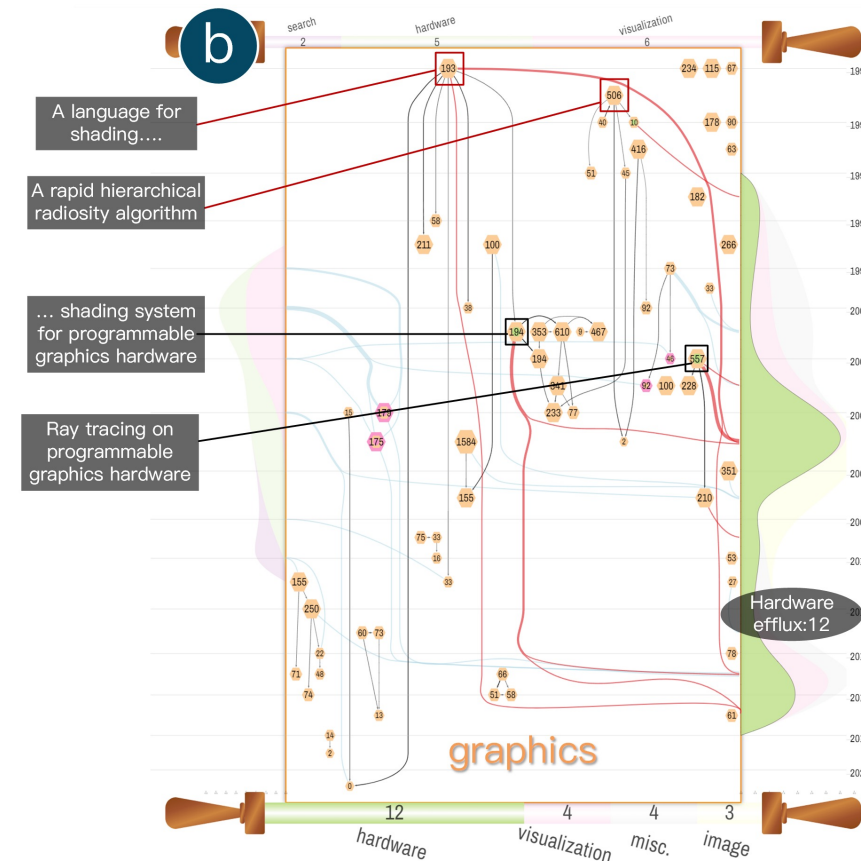
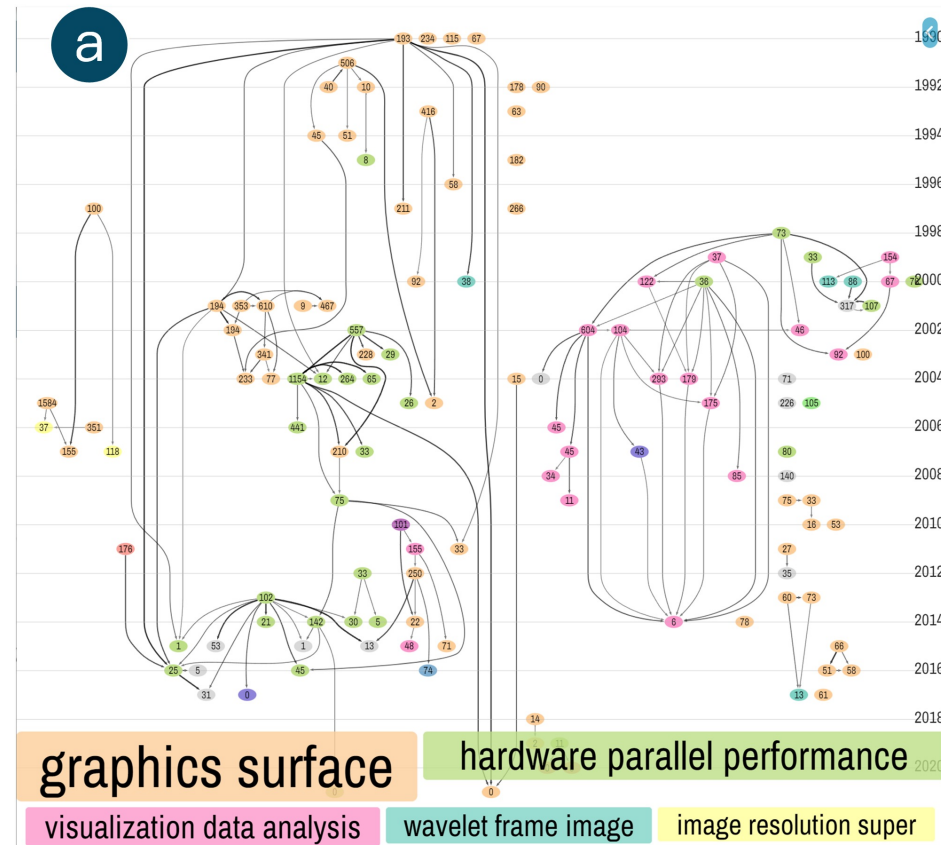
Category	Name	#Authors	#Topic	#Papers	#Links
Award Recipients	ACM Fellows	1285	324	421205	818548
	Turing Award Winners	67	324	17625	30780
Top Scholars in 12 CS Sub-fields	Graphics & Visualization	11980	105	271418	262957
	Artificial Intelligence (Full list in Appendix)	54637	497	4836561	8420229
Academic Venues	International Symposium on Graph Drawing	N/A	28	1703	3128

GF Graph Statistics



Case Study

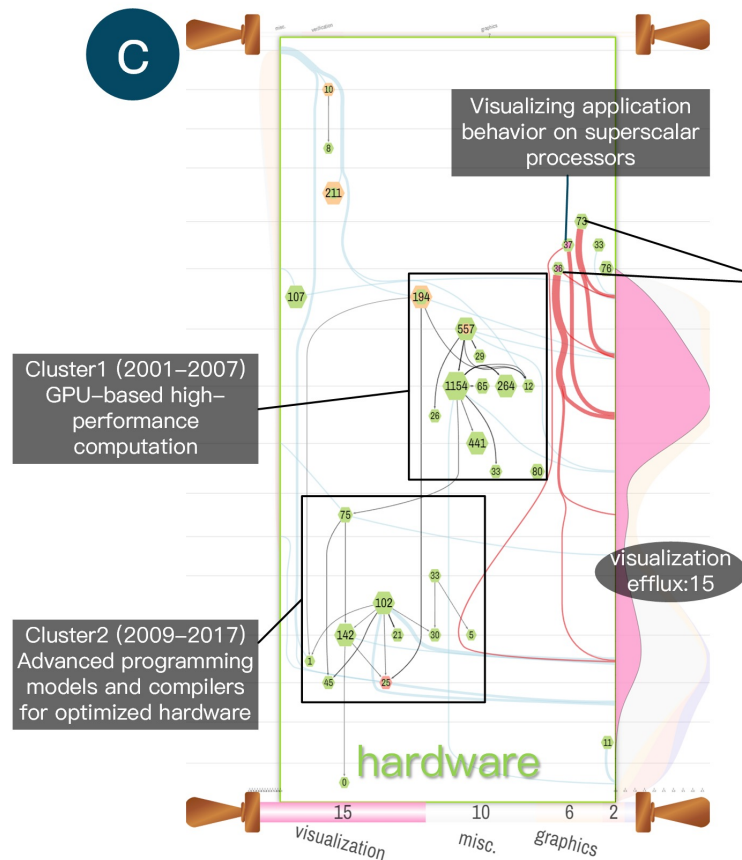
Prof. Hanrahan's Interdisciplinary Impact



- Three primary research topics: 3D graphics (orange), parallel hardware (green), and visualization (pink)
- 3D graphics topic:
 - two pioneering papers (RenderMan, hierarchical radiosity) form two main research threads
 - gradually incubated another topic (Parallel Hardware), featured by two interdisciplinary papers

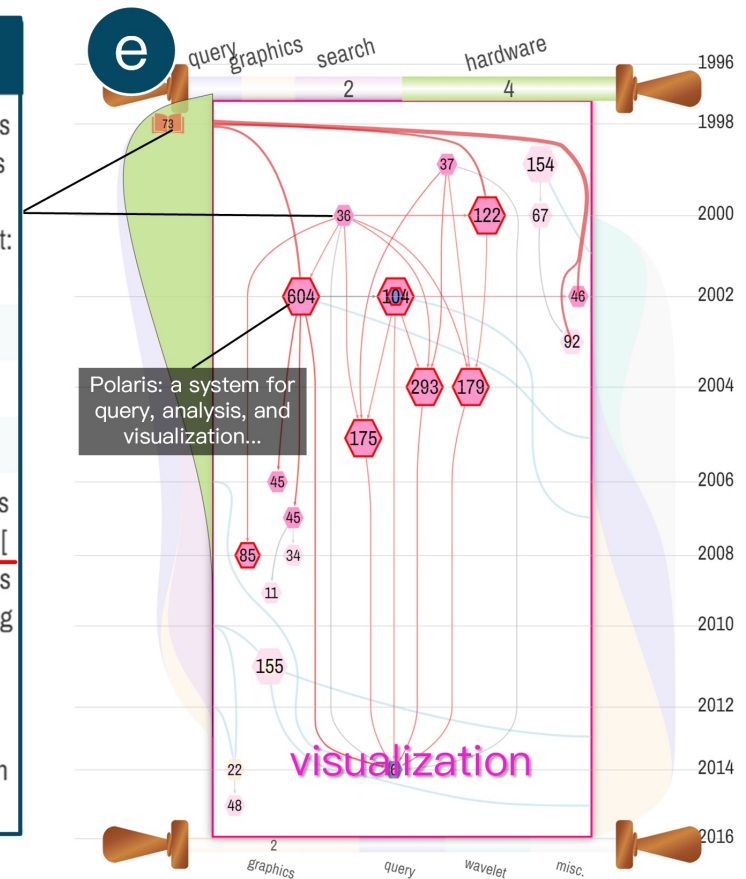
Case Study

Prof. Hanrahan's Interdisciplinary Impact



d Citation Relationship

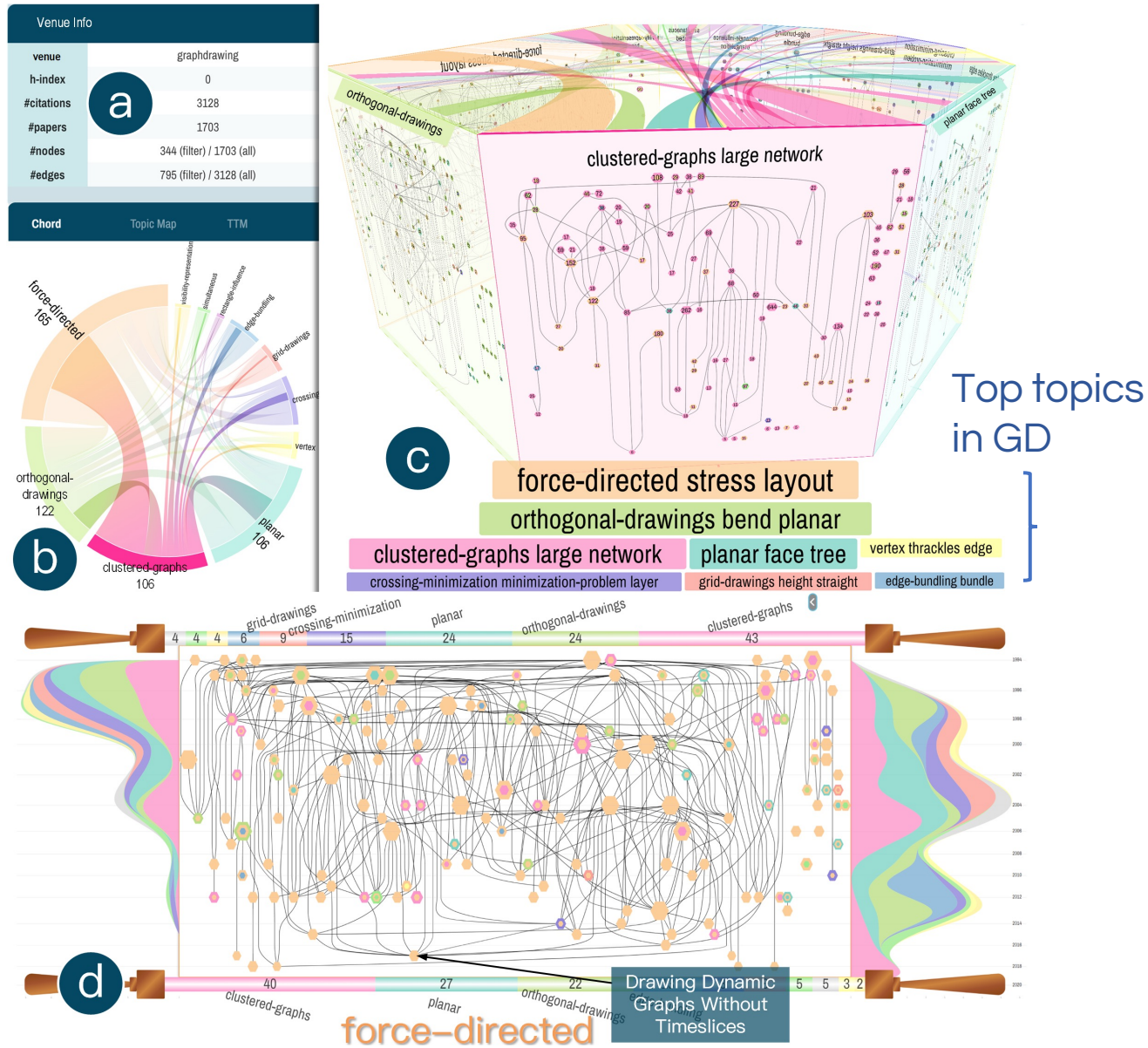
cited paper	The design of a parallel graphics interface, Performance analysis and visualization of title parallel systems using SimOS and Rivet: a case study
year	1998, 2000
venue	SIGGRAPH, HPCA
#citation	73, 36
citation context	7.2 Scenario 2: Computer Systems Analysis At Stanford, researchers developing Argus [21], a parallel graphics library, found that its performance had linear speedup when using up to 31 processors, after which its performance diminished rapidly. // Using <u>Polaris</u> , we recreate the analysis they performed using a custom-built visualization tool[8].



- Parallel hardware topic:
 - two sub-topics (Cluster1/2), again triggers a new topic “visualization”
 - reveal a key clue on how they developed the famous Polaris visualization system
- Visualization topic: primary component is seven publications focusing on Polaris

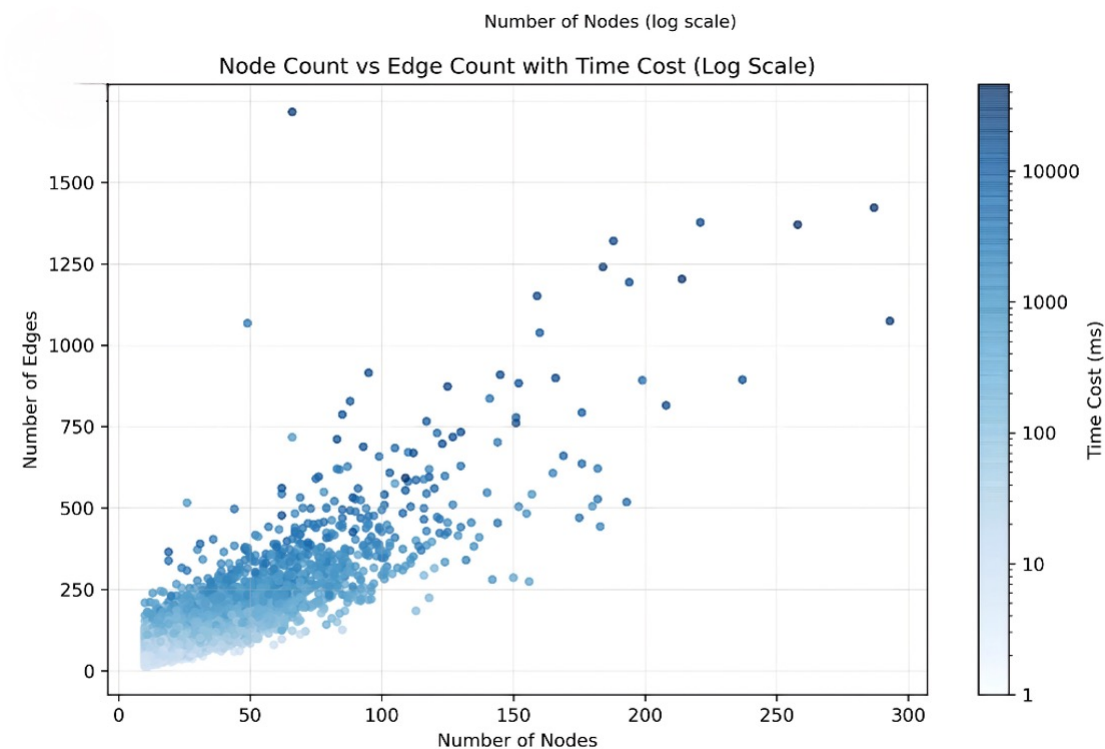
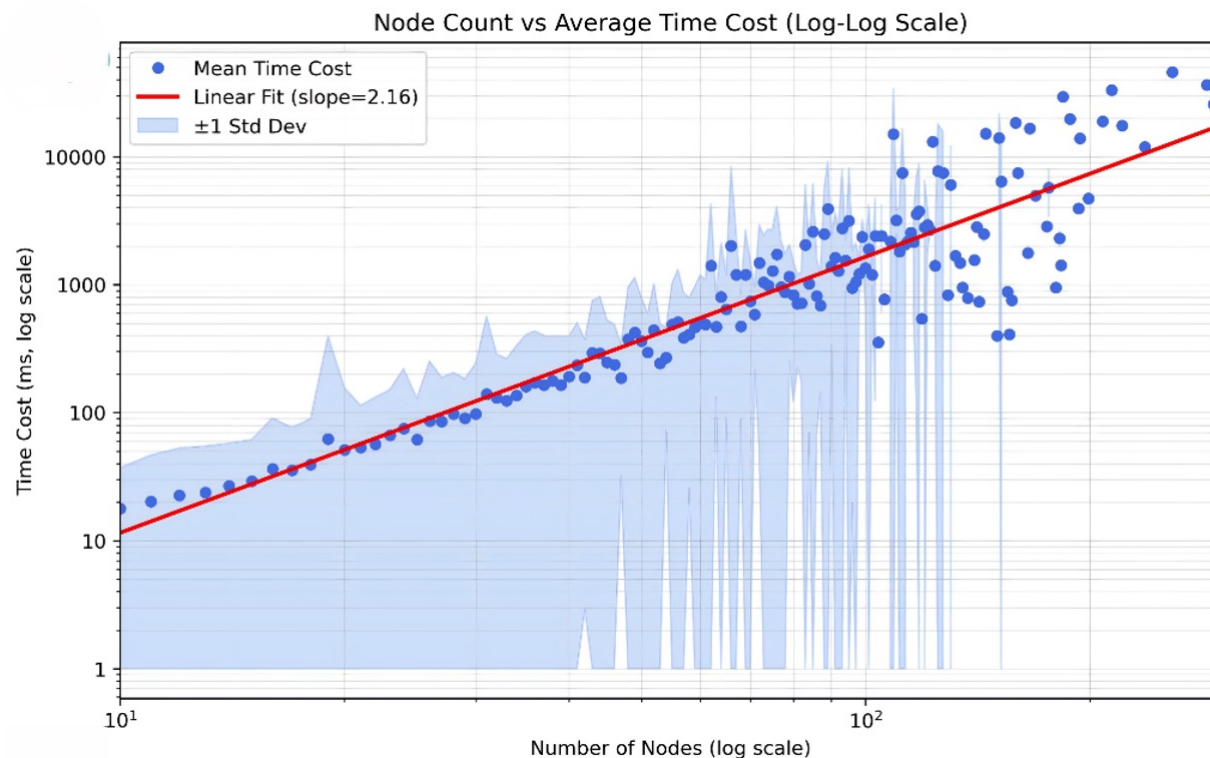
Case Study

The Backbone and Outreach of Graph Drawing Symposium



- All the GD papers from MAG (1994~2021), high-impact GD papers (344/1703) form 11 topics
- Force-directed topic:
 - More than half encoded with multiple color layers, representing interdisciplinary research applying to various problems
 - Highest-impact papers are single-colored, indicating major scientific evolution

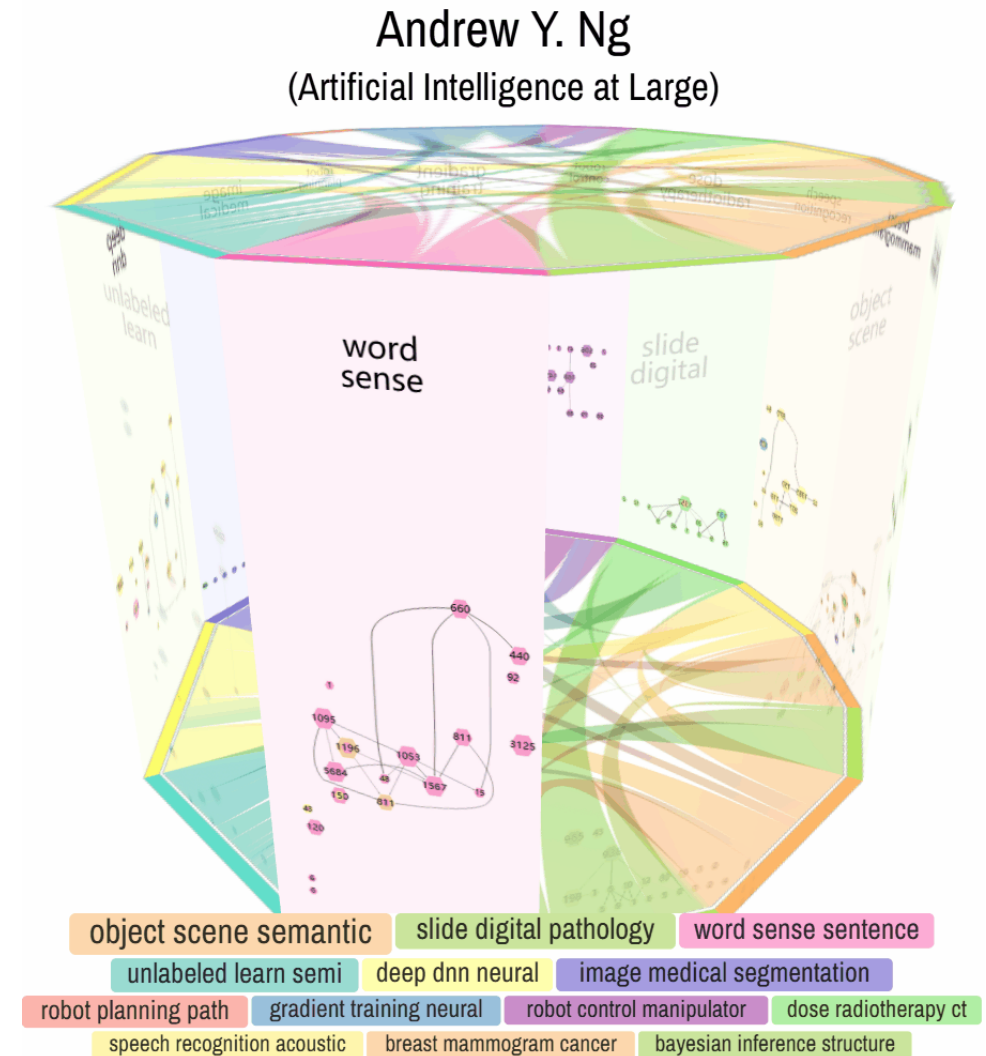
Complexity & Efficiency Analysis



- Complexity of barycenter methods (similar to Graphviz DOT) ranging from $O(|V||E|)$ to $O(|V|^3)$
- Experiment on 20k small subgraphs ($|V| \geq 10$) showed quadratic complexity in practice ($O(|V|^2)$)
- Our method decomposes the large GF graph into per-topic GF subgraph, ensuring interactive rendering

Conclusion & Future Work

- Contribution:
 - Solve the problem of visualizing the research evolution of individual scholars from multiple topic facets
 - Introduce **GeneticPrism**, a novel 3D prism shaped visual metaphor, to integrate multiple topic facets
 - Introduce **GeneticScroll and FiHL algorithm** for intra-topic research evolution with topic-level interactions
- Future Work:
 - GeneticPrism design can be readily extended to visualize generic **multivariate networks** with a primary node attribute working as the topic attribute
 - Extending GeneticPrism to support the analysis of **multiple scholars** in the same group or community





The 33rd International Symposium on Graph Drawing and Network Visualization

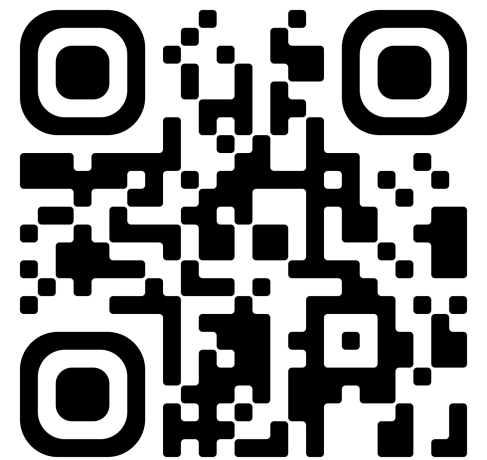
Thank you for your attention !

Github Repository: <https://github.com/visdata/GeneticPrism>

Contact: geneticflowcontact@gmail.com



北京航空航天大学
BEIHANG UNIVERSITY

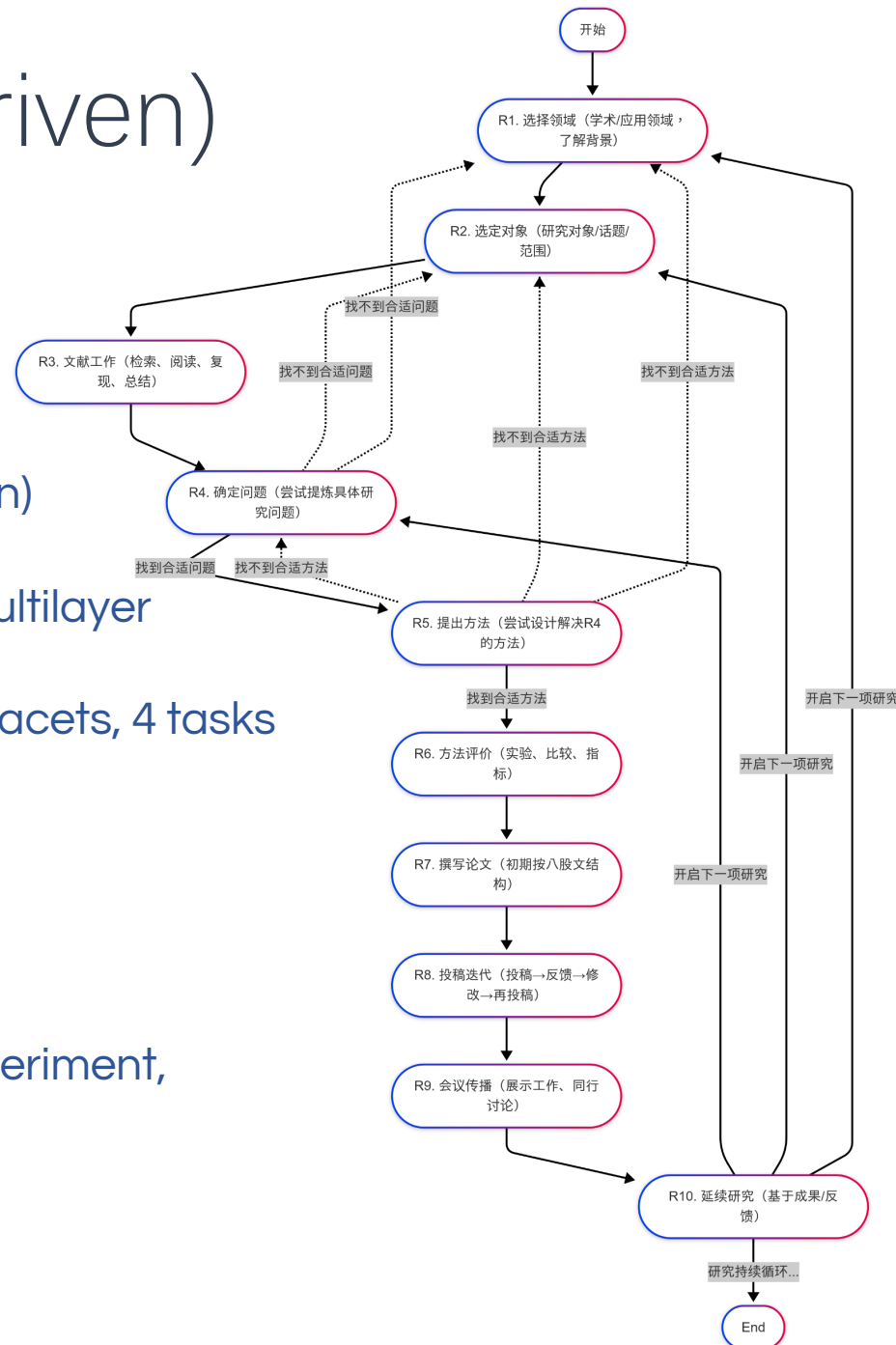


<https://genetic-flow.com>

The Research Pipeline (Paper-Driven)

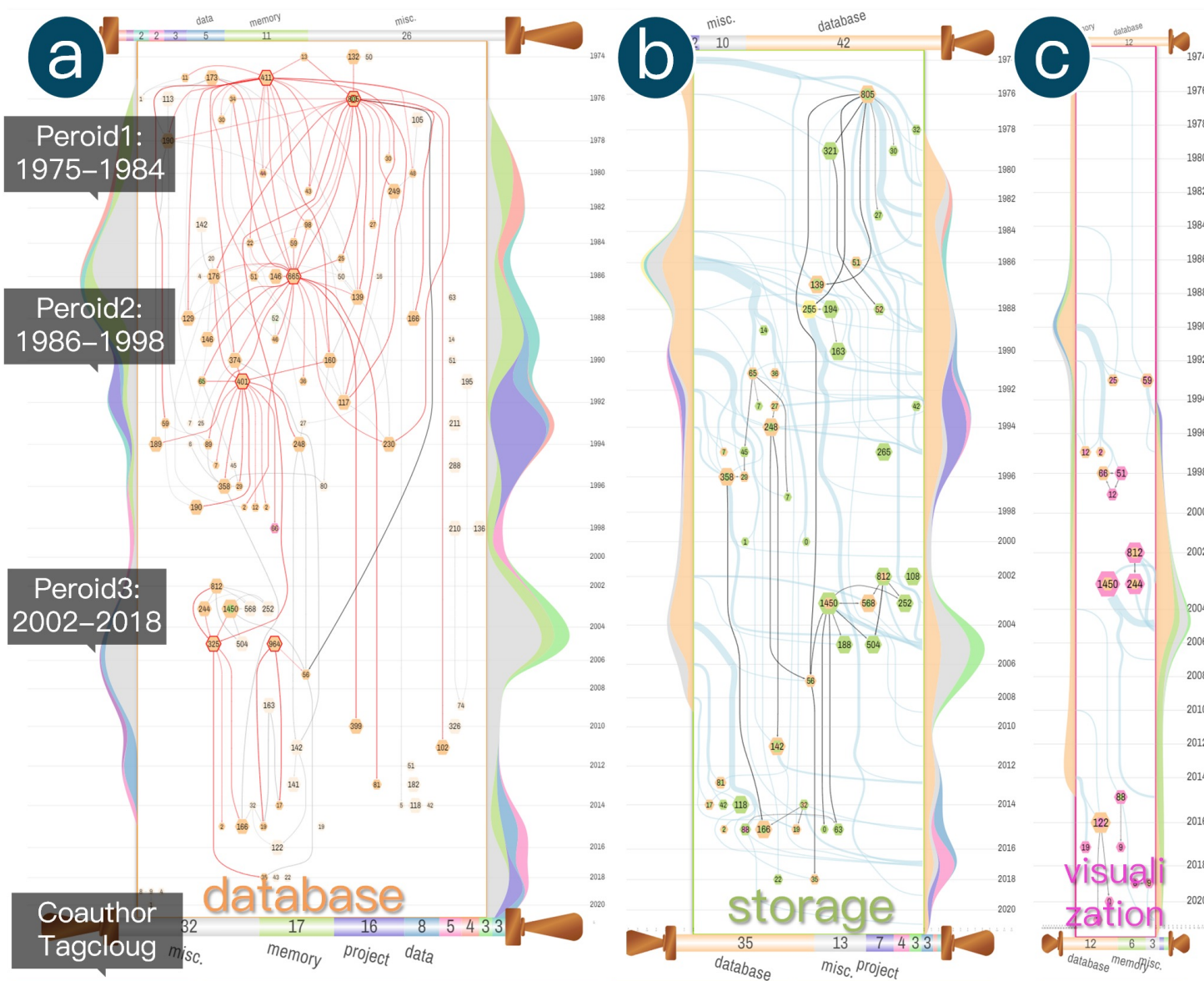
每个阶段干专门的事

- R1. 选择领域 Information Visualization (for science)
 - R2. 选择特定研究对象 Academic Network (self-citation)
 - R3. 分析相关工作(文献) Scholarly Research Data & multilayer
 - R4. 确定具体研究问题 Visualization for multiple topic facets, 4 tasks
 - R5. 提出解决R4问题的方法 topic-based GF graph
 - Central & Context
 - Overview & Detailed
 - Algorithm & Design
 - R6. 评价R5方法的有效性
 - R7. 撰写论文 Case study, algorithm experiment, user evaluation
 - R8. 投稿、修改、再投稿
 - R9. 会议present



Case Study

Prof. Stonebraker's Lifelong Database Research



- Database topic dominates his research with 107 core papers out of all 174 papers (61.5%)
- Identifies three major periods
 - INGRES (1975-1984)
 - Postgres (1986-1998)
 - Commercial DBMS (2002-)
- Other topics contain widely distributed interdisciplinary research centered on “database” topic